Maxwell Equation in Dielectric Materials

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2026 年 4 月 26 日

1 The Maxwell Equation

在 [14] 中，处理了带有周期边界条件下的分数阶 Maxwell 方程组，相应的方程为：

$$\begin{aligned} \mu_0 \frac{\partial H}{\partial t} &= \frac{\partial E}{\partial x}, && \text{in } \Omega \times (0,T] \\ \epsilon_0 \epsilon_\infty \frac{\partial E}{\partial t} &= \frac{\partial H}{\partial x} - \frac{\partial P}{\partial t}, && \text{in } \Omega \times (0,T] \\ \tau_0^\alpha \frac{\partial^\alpha P}{\partial t^\alpha} + P &= \varepsilon_0 \left(\varepsilon_s - \varepsilon_\infty\right) E, && \text{in } \Omega \times (0,T] \end{aligned} \tag{1.1}$$

old maxwell equ

这里给出的是 1 维情况下，不带 UPML 层的方程。这个方程的好处就是将电极化强度 P 直接入了方程组中，而我们之前做的方法是在原有的简单 Maxwell 方程中带入电极化强度，然后再在化简的过程中分理处分数阶的方程，较为麻烦。

转而使用这样的方程形式，在这个方程组的基础上添加 UPML 层。

但这个方程在归一化后等价于卢老师文章中所写的 Maxwell Equation

1.1 dielectric response

对于一般的媒质，本构关系为：

$$\begin{aligned} \mathbf{D} &= \epsilon_0 \epsilon_{r,\infty} \mathbf{E} + \mathbf{P} \\ \mathbf{B} &= \mu_0 \mu_{r,\infty} \mathbf{H} + \mathbf{M} \end{aligned} \tag{1.2}$$

其中 ε_0, μ_0 分别表示真空下的电容率，导磁率；$\varepsilon_{r,\infty}, \mu_{r,\infty}$ 分别表示无穷频率下的相对电容率，相对导磁率。$\mathbf{B}$ 为磁感应强度，$\mathbf{D}$ 为电位移，$\mathbf{P}$ 为电极化强度，$\mathbf{M}$ 为磁极化强度。

一般来说, 在色散介质中，电极化强度 P 的分量 P_i 和电场强度 $\mathbf{E}$ 的分量的关系在频域 (用 $\breve{}$ 表示) 可以表示幂级数的形式,

$$\breve{P}_i = \epsilon_0 \sum_j \chi_{ij}(\omega)\breve{E}_j + \epsilon_0^2 \sum_{j,k} \chi_{ijk}(\omega)\breve{E}_j\breve{E}_k + O\left(\breve{E}^3\right).$$

对于线性，各向同性的介质，可以只保留首项:

$$\mathbf{P} = \epsilon_0 \chi(\omega)\mathbf{E} \tag{1.3}$$

其中 $\chi(\omega)$ 为矩阵，称作这个媒质上的频域极化率。

频域中实验介电响应的精确表示通常不能用具有单个弛豫时间的简单指数表达式（Debye 模型）来描述。对于 Debye 介质，其在频域上的极化率为:

$$\chi(\omega) = \frac{\epsilon_{r,s} - \epsilon_{r,\infty}}{1 + j\omega\tau} \tag{1.4}$$

其中 $\varepsilon_{r,s}$ 为静态相对电容率（静态电场下的 ε，在静态或者低频的条件下，物质的极化能够充分响应电场的变化），τ 为弛豫时间（用来描述系统达到热动平衡所需的时间）。

而这种模型的精确度较低，有许多其他的经验关系，包括 Cole–Cole（C–C）、Cole–Davidson（C–D）和 Havriliak–Negami（H–N）方程，用来拟合 dielectric spectra。

这样使得介电响应的模型更加灵活，使得能够更好地参数化色散介质特性。

对于频域上 H-N 方程描述的复介电常数为:

$$\varepsilon_r(\omega) = \varepsilon_{r,\infty} + \frac{\varepsilon_{r,s} - \varepsilon_{r,\infty}}{\left[1 + (j\omega\tau)^{1-\alpha}\right]^{1-\beta}} \tag{1.5}$$

其中 $\varepsilon_{r_s}, \varepsilon_{r_\infty}$ 分别表示 static and infinite relative permittivity, τ 表示弛豫时间, $\alpha, \beta \in [0, 1]$ 表示介电常数的形状和宽度。

当 $\alpha = \beta = 0$ 时，描述的就是 Debye 介质；当 $\alpha \in [0, 1], \beta = 0$ 时，描述的是 C-C 方程；当 $\alpha = 0, \beta \in [0, 1]$ 时，描述的是 C-D 方程。

更一般的，有 Raicu 方程进行描述，即上述的 H-N 方程的分母为 $[(j\omega\tau)^v + (j\omega\tau)^\alpha]^\beta$

则电极化强度 $\mathbf{P}$ 和电场强度 $\mathbf{E}$ 之间的关系式可以写做:

$$\mathbf{P} = \epsilon_0(\epsilon_r - \epsilon_{r,\infty})\mathbf{E}. \tag{1.6}$$

通过 Laplace 逆变换后，对应式(1.1) 的第三式。

1.2 Usual Maxwell Equation

一般的 Maxwell 旋度方程为

$$\begin{aligned} \nabla \times \boldsymbol{H} &= \frac{\partial \boldsymbol{D}}{\partial t} + \boldsymbol{J} \\ \nabla \times \boldsymbol{E} &= -\frac{\partial \boldsymbol{B}}{\partial t} - \boldsymbol{J}_m \end{aligned} \tag{1.7}$$

其中：$\boldsymbol{E}$ 为电场强度，单位为伏特 / 米 (V/m)；D 为电通量密度, 单位为库仑 / 米 ^{2} $\left(\mathrm{C/m^2}\right)$；$\boldsymbol{H}$ 为磁场强度，单位为安培 / 米 (A/m)；B 为磁通量密度，单位为韦伯 / 米 ^{2} $\left(\mathrm{Wb/m^2}\right)$；$J$ 为电流密度, 单位为安培/ 米 ^{2} $\left(\mathrm{A/m^2}\right)$；$J_m$ 为磁流密度, 单位为伏特 / 米 ^{2} $\left(\mathrm{V/m^2}\right)$。

对于 1 维情形下，$\frac{\partial}{\partial y} = 0, \frac{\partial}{\partial z} = 0$，带入三维情况下的方程计算，使得其方程两端的符号确实和 3 维情况下不同。（TEM 波分为两组表达形式，仅仅是空间导数项存在一个 $\pm$ 的差别，影响不大）

1.3 一些介质

[2, 4] 中提到了一些介质的模型。我们称一些介质是色散的，如果他们在频域空间中，不是简单的依赖于介电常数 ε 和磁导率 μ。特别的，如果 $\mu(s) = \epsilon(s) = 1, s \in \mathbb{C}^+$，那么就称这个介质是非色散的。我们考虑的介质都需要满足以下假设。假设 $\eta : \mathbb{C}_\star^+ \to \mathbb{C}$ 表示 ε 或者 μ，那么我们称一个 $\mathbb{C}_\star^+$ 上的解析函数是

- **实值 (time-real)**，如果 $\eta(\bar{s}) = \bar{\eta(s)}, s \in \mathbb{C}_\star^+$. 这个条件是为了说明 $\mathcal{L}^{-1}\eta(s)$ 是实值的因果分布 (causal distribution)。
- **被动 (passive)**，如果 $\mathrm{Re}(s\eta(s)) > 0, s \in \mathbb{C}_\star^+$。这个条件和问题的稳定性相关 [4]。
- **在高频下非色散 (non-dispersive at high frequencies)**，如果对于所有的 $\alpha \in (-\frac{\pi}{2}, \frac{\pi}{2})$, $\lim_{R\to+\infty} \eta(Re^{i\alpha}) = 1$.

上述的这些定理都已经 [4] 中出现过，只是在最后一点高频下的非色散的表述不太一样。

Definition 1.1. 称一个函数 $\eta : \mathbb{C}_\star^+ \to \mathbb{C}$ 是 **相容的** ***(admissible)***，如果 η 在 $\mathbb{C}_\star^+$ 上是解析的，并且满足上述的三个条件。

接下的所有讨论的介质都是相容的。

Assumption 1.1. 假设函数 $s \mapsto \varepsilon(s)$ 和 $s \mapsto \mu(s)$ 是相容的。

2 The Maxwell Equation with UPML

ation with UPML

在前面的章节中给出了周期边界条件下分数阶 Maxwell 方程组的表示形式，现在要给出在 UPML 吸收边界条件下的分数阶 Maxwell 方程组。

卢老师论文中，构造的 Maxwell 方程将电极化强度 $\mathbf{P}$ 这项用 $\mathbf{E}$ 这项表示，写入整体之中，现在不做合并，将 $\mathbf{P}$ 保留，得到：

$$\begin{aligned} \nabla \times \breve{\mathbf{H}} &= \left(j\omega\epsilon_0\epsilon_{r,\infty}\breve{\mathbf{E}} + j\omega\breve{\mathbf{P}} + \sigma\breve{\mathbf{E}} \right) \bar{\bar{\epsilon}}, \\ \nabla \times \breve{\mathbf{E}} &= -j\omega\mu_0\mu_r\bar{\bar{\mu}}\breve{\mathbf{H}}, \\ \left[1 + (jw\tau)^{\alpha}\right] \breve{\mathbf{P}} &= \epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty})\breve{\mathbf{E}}. \end{aligned} \tag{2.1}$$

Maxwell in C-C

其中

$$\bar{\bar{\epsilon}} = \bar{\mu} = \begin{bmatrix} \frac{s_y s_z}{s_x} & 0 & 0 \\ 0 & \frac{s_x s_z}{s_y} & 0 \\ 0 & 0 & \frac{s_x s_y}{s_z} \end{bmatrix} \tag{2.2}$$

且

$$s_i = 1 + \frac{\sigma_i}{j\omega\epsilon_0}, \quad i = x, y, z \tag{2.3}$$

ε_r 表示色散介质的相对电容率，其中 σ_i 作为 UPML 层的相关参数，当 $\sigma_i = 0$ 对应于物理的色散介质区域，先考虑 Cole-Cole 介质作为相对电容率。

2.1 parameter settings for UPML

先给出反射系数，再推导 $\sigma_{\max}$ 的取值。

在连续空间下，如果一个波以角度 θ 入射 PML 层，法线方向为 w，其吸收因子记为：

$$R(\theta) = e^{-2\sigma_w \eta d \cos\theta}, \tag{2.4}$$

其中 η 为阻抗，σ_w 为 w 方向的衰减系数。

一般的，将 UPML 层中的电导 σ_i 设计为随多项式变化的形式：

$$\sigma_i(x) = (l/d)^m \sigma_{i,\max} \tag{2.5}$$

其中 l 是到计算区域和PML 之间的交界面的距离, d 是PML 层的厚度。从上式中可以看出, σ_i 在 UPML 层的内边界 $l=0$ 是零, 在PML 的外边界 $l=d$ 增加到最大值 $\sigma_{i,\max}$ 。在 PML 外边界上使用 PEC 边界条件,UPML 层的反射因子取做:

$$R(\theta)=e^{-2\eta\cos\theta\int_0^d\sigma_i(x)dx}, \tag{2.6}$$

再进行积分得到:

$$R(\theta)=e^{-2\eta\sigma_{i,\max}d\varepsilon_{r,\infty}\cos\theta/(m+1)}=e^{-2\sigma_{i,\max}d\cos\theta/v(m+1)} \tag{2.7}$$

标红的部分与 Taflove 中不同 其中 η 是PML 的阻抗, θ 是入射角, $v=(\eta\epsilon_{r,\infty})^{-1}=(\epsilon_{r,\infty}\mu_r)^{-1/2}$ 是光在该物质中的速度。我们可以选取适当的PML 参数来满足对于给定的垂直反射射因子。例如, 已知 m,d 和垂直反射因子 $R(0),\sigma_{i,\max}$ 可以用如下公式计算

$$\sigma_{i,\max}=-\frac{v(m+1)\ln(R(0))}{2d}$$

而 Taflove 中取法为

$$\sigma_{i,\max}=-\frac{(m+1)\ln(R(0))}{2\eta d} \tag{2.8}$$

其中

$$\eta=\sqrt{\frac{\mu}{\epsilon}}=\sqrt{\frac{\mu_0\mu_r}{\epsilon_0\epsilon_r}}=\sqrt{\frac{\mu_r}{\epsilon_{r,\infty}}}$$

归一化后，η 取值始终为有限值。对于 10 层 PML 网格，一般取反射因子 $R(0)=e^{-16}$. 在数值算例中 [13, 14]，参数的选取为

$$\begin{cases}\mu_0=\tau_0=\epsilon_0=\epsilon_\infty=1,\epsilon_s=2,\\ \mu_0=\tau_0=\epsilon_0\epsilon_\infty=\epsilon_0\left(\epsilon_s-\epsilon_\infty\right)=1\end{cases}$$

2.2 2-D Maxwell Equation with UPML in Cole-Cole dispersive medium

简单起见，先考虑2维TM波的情况($\sigma_z = 0$)，在频域下的的Maxwell Equation在Cole-Cole介质下写做：

$$\frac{\partial \breve{H}_y}{\partial x} - \frac{\partial \breve{H}_x}{\partial y} = \left(j\omega\epsilon_0\epsilon_{r,\infty}\breve{E}_z + j\omega\breve{P} + \sigma\breve{E}_z\right) s_x s_y, \tag{2.9a}$$ primal Maxwell

$$\frac{\partial \breve{E}_z}{\partial y} = -j\omega\mu_0\mu_r \frac{s_y}{s_x}\breve{H}_x, \tag{2.9b}$$ primal Maxwell

$$-\frac{\partial \breve{E}_z}{\partial x} = -j\omega\mu_0\mu_r \frac{s_x}{s_y}\breve{H}_y, \tag{2.9c}$$ primal Maxwell

$$P + \tau^\alpha \frac{\partial^\alpha P}{\partial t^\alpha} = \epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty})\breve{E}_z. \tag{2.9d}$$ primal Maxwell

将方程 (2.9a)写做：

$$\frac{\partial \breve{H}_y}{\partial x} - \frac{\partial \breve{H}_x}{\partial y} = j\omega\epsilon_0\epsilon_{r,\infty}\breve{E}_z + \sigma\breve{E}_z + j\omega\breve{P} + \breve{J_{z,1}}(\omega) + \breve{J_{z,2}}(\omega) + \breve{J_{z,3}}(\omega) \tag{2.10}$$ J style Maxwell

其中

$$\begin{aligned}
\breve{J_{z,1}}(\omega) &= j\omega\epsilon_0\epsilon_{r,\infty}(s_x s_y - 1)\breve{E}_z \\
\breve{J_{z,2}}(\omega) &= j\omega(s_x s_y - 1)\breve{P} \\
\breve{J_{z,3}}(\omega) &= \frac{\sigma}{j\omega\epsilon_0\epsilon_{r,\infty}}\breve{J}_{z,1}(\omega)
\end{aligned} \tag{2.11}$$

带入 s_x, s_y 的表达式，对于 $\breve{J}_{z,1}, \breve{J}_{z,2}$ 得到

$$\begin{aligned}
\breve{J}_{z,1}(\omega) &= j\omega\epsilon_0\epsilon_{r,\infty}\left(\left(1 + \frac{\sigma_x}{j\omega\epsilon_0}\right)\left(1 + \frac{\sigma_y}{j\omega\epsilon_0}\right) - 1\right)\breve{E}_z \\
&= j\omega\epsilon_0\epsilon_{r,\infty}\left(\frac{\sigma_x + \sigma_y}{j\omega\epsilon_0} + \frac{\sigma_x\sigma_y}{(j\omega\epsilon_0)^2}\right)\breve{E}_z \\
&= \epsilon_{r,\infty}\left(\sigma_x + \sigma_y\right)\breve{E}_z + \frac{\epsilon_{r,\infty}}{j\omega\epsilon_0}\sigma_x\sigma_y\breve{E}_z,
\end{aligned} \tag{2.12}$$

$$\begin{aligned}
\breve{J}_{z,2}(\omega) &= j\omega\left(\left(1 + \frac{\sigma_x}{j\omega\epsilon_0}\right)\left(1 + \frac{\sigma_y}{j\omega\epsilon_0}\right) - 1\right)\breve{P} \\
&= j\omega\left(\frac{\sigma_x + \sigma_y}{j\omega\epsilon_0} + \frac{\sigma_x\sigma_y}{(j\omega\epsilon_0)^2}\right)\breve{P} \\
&= \frac{\sigma_x + \sigma_y}{\epsilon_0}\breve{P} + \frac{\sigma_x\sigma_y}{j\omega\epsilon_0^2}\breve{P},
\end{aligned} \tag{2.13}$$

通过 Laplace 逆变换的方式，给出对于 $J_{z,1}, J_{z,2}, J_{z,3}$ 的控制方程。

对于 $\breve{J}_{z,1}$，两边同乘 $j\omega$，Laplace 逆变换得到：

$$\frac{\partial J_{z,1}(t)}{\partial t} = \epsilon_{r,\infty}(\sigma_x + \sigma_y)\frac{\partial E_z}{\partial t} + \frac{\epsilon_{r,\infty}}{\epsilon_0}\sigma_x\sigma_y E_z, \tag{2.14}$$

引入变量 $P_{z,1} = -J_{z,1} + \epsilon_{r,\infty}(\sigma_x + \sigma_y)E_z$，上式化简为：

$$\frac{\partial P_{z,1}}{\partial t} = -\frac{\epsilon_{r,\infty}}{\epsilon_0}\sigma_x\sigma_y E_z. \tag{2.15}$$

对于 $\breve{J}_{z,2}$，两边同乘 $j\omega$，Laplace 逆变换得到：

$$\frac{\partial J_{z,2}(t)}{\partial t} = \frac{\sigma_x + \sigma_y}{\epsilon_0}\frac{\partial P}{\partial t} + \frac{\sigma_x\sigma_y}{\epsilon_0^2}P, \tag{2.16}$$

引入变量 $P_{z,2} = -J_{z,2} + \frac{\sigma_x+\sigma_y}{\epsilon_0}P$，上式化简为：

$$\frac{\partial P_{z,2}}{\partial t} = -\frac{\sigma_x\sigma_y}{\epsilon_0^2}P. \tag{2.17}$$

对于 $\breve{J}_{z,3}$，两边同乘 $j\omega$，Laplace 逆变换得到：

$$\frac{\partial J_{z,3}(t)}{\partial t} = \frac{\sigma}{\epsilon_0\epsilon_{r,\infty}}J_{z,1}, \tag{2.18}$$

带入 $J_{z,1}$ 的表达式，并为了方便起见，令 $P_{z,3} = -J_{z,3}$，上式化简为：

$$\frac{\partial P_{z,3}(t)}{\partial t} = \frac{\sigma}{\epsilon_0\epsilon_{r,\infty}}P_{z,1} - \frac{\sigma(\sigma_x + \sigma_y)}{\epsilon_0}E_z. \tag{2.19}$$

对方程(2.10)采用 Laplace 逆变化，得到等价的时域上的方程：

$$\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = \epsilon_0\epsilon_{r,\infty}\frac{\partial E_z}{\partial t} + \sigma E_z + \frac{\partial P}{\partial t} + J_{z,1}(t) + J_{z,2}(t) + J_{z,3}(t). \tag{2.20}$$

带入 $J_{z,1}, J_{z,2}, J_{z,3}$ 的表达式，得到：

$$\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = \epsilon_0\epsilon_{r,\infty}\frac{\partial E_z}{\partial t} + \sigma E_z + \frac{\partial P}{\partial t} - P_{z,1} + \epsilon_{r,\infty}(\sigma_x + \sigma_y)E_z - P_{z,2} + \frac{\sigma_x + \sigma_y}{\epsilon_0}P - P_{z,3}. \tag{2.21}$$

对于方程 (2.9b)，通过同样的方式，可以得到如下的推导：

$$
\begin{aligned}
\frac{\partial \breve{E}_z}{\partial y} &= -j\omega\mu_0\mu_r \frac{j\omega\epsilon_0+\sigma_y}{j\omega\epsilon_0+\sigma_x}\breve{H}_x \\
&= -j\omega\mu_0\mu_r(1+\frac{\sigma_y-\sigma_x}{j\omega\epsilon_0+\sigma_x})\breve{H}_x \\
&= -j\omega\mu_0\mu_r\breve{H}_x - j\omega\mu_0\mu_r\frac{\sigma_y-\sigma_x}{j\omega\epsilon_0+\sigma_x}\breve{H}_x \\
&= -j\omega\mu_0\mu_r\breve{H}_x - j\omega\mu_0\mu_r(\frac{\sigma_y-\sigma_x}{j\omega\epsilon_0} - \frac{(\sigma_y-\sigma_x)\sigma_x}{j\omega\epsilon_0(j\omega\epsilon_0+\sigma_x)})\breve{H}_x
\end{aligned} \tag{2.22}
$$

令

$$
\breve{Q}_x = \mu_0\mu_r\frac{(\sigma_y-\sigma_x)\sigma_x}{\epsilon_0(j\omega\epsilon_0+\sigma_x)}\breve{H}_x \tag{2.23}
$$

Q_x definition

那么上式可以得到：

$$
\frac{\partial \breve{E}_z}{\partial y} = -j\omega\mu_0\mu_r\breve{H}_x - \mu_0\mu_r\frac{\sigma_y-\sigma_x}{\epsilon_0}\breve{H}_x + \breve{Q}_x \tag{2.24}
$$

再通过 Fourier 逆变换，得到：

$$
\frac{\partial E_z}{\partial y} = -\mu_0\mu_r\frac{\partial H_x}{\partial t} - \mu_0\mu_r\frac{\sigma_y-\sigma_x}{\epsilon_0}H_x + Q_x \tag{2.25}
$$

通过方程(2.23) 可以得到 Q_x 的控制方程：

$$
\begin{aligned}
\frac{j\omega\epsilon_0+\sigma_x}{\epsilon_0}\breve{Q}_x &= \mu_0\mu_r\frac{(\sigma_y-\sigma_x)\sigma_x}{\epsilon_0^2}\breve{H}_x \\
j\omega\breve{Q}_x &= -\frac{\sigma_x}{\epsilon_0}\breve{Q}_x + \mu_0\mu_r\frac{(\sigma_y-\sigma_x)\sigma_x}{\epsilon_0^2}\breve{H}_x
\end{aligned}
$$

通过 Fourier 逆变换得到：

$$
\frac{\partial Q_x}{\partial t} = -\frac{\sigma_x}{\epsilon_0}Q_x + \mu_0\mu_r\frac{(\sigma_y-\sigma_x)\sigma_x}{\epsilon_0^2}H_x \tag{2.26}
$$

那么对于 Faraday 定律的方程组 (2.9b),(2.9c)，得到如下的表达:

$$
\begin{aligned}
\frac{\partial E_z}{\partial y} &= -\mu_0\mu_r\frac{\partial H_x}{\partial t} - \mu_0\mu_r\frac{\sigma_y-\sigma_x}{\epsilon_0}H_x + Q_x, \\
-\frac{\partial E_z}{\partial x} &= -\mu_0\mu_r\frac{\partial H_y}{\partial t} - \mu_0\mu_r\frac{\sigma_x-\sigma_y}{\epsilon_0}H_y + Q_y, \\
\frac{\partial Q_x}{\partial t} &= -\frac{\sigma_x}{\epsilon_0}Q_x + \frac{\mu_0\mu_r\sigma_x\left(\sigma_y-\sigma_x\right)}{\epsilon_0^2}H_x, \\
\frac{\partial Q_y}{\partial t} &= -\frac{\sigma_y}{\epsilon_0}Q_y + \frac{\mu_0\mu_r\sigma_y\left(\sigma_x-\sigma_y\right)}{\epsilon_0^2}H_y.
\end{aligned} \tag{2.27}
$$

对上述的推导我们可以得到对于 $H_x, H_y, E_z, P, P_{z,1}, P_{z,2}, P_{z,3}, Q_x, Q_y$ 为变量的方程组：

$$\epsilon_0\epsilon_{r,\infty}\frac{\partial E_z}{\partial t} = \frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} - \left[\sigma + \epsilon_{r,\infty}(\sigma_x + \sigma_y)\right] E_z - \frac{\partial P}{\partial t} - \frac{\sigma_x + \sigma_y}{\epsilon_0} P + P_{z,1} + P_{z,2} + P_{z,3}, \tag{2.28a}$$

$$\mu_0\mu_r\frac{\partial H_x}{\partial t} = -\frac{\partial E_z}{\partial y} - \mu_0\mu_r\frac{\sigma_y - \sigma_x}{\epsilon_0} H_x + Q_x, \tag{2.28b}$$

$$\mu_0\mu_r\frac{\partial H_y}{\partial t} = \frac{\partial E_z}{\partial x} - \mu_0\mu_r\frac{\sigma_x - \sigma_y}{\epsilon_0} H_y + Q_y, \tag{2.28c}$$

$$P + \tau^\alpha\frac{\partial^\alpha P}{\partial t^\alpha} = \epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty})E_z, \tag{2.28d}$$

$$\frac{\partial P_{z,1}}{\partial t} = -\frac{\epsilon_{r,\infty}}{\epsilon_0}\sigma_x\sigma_y E_z, \tag{2.28e}$$

$$\frac{\partial P_{z,2}}{\partial t} = -\frac{\sigma_x\sigma_y}{\epsilon_0^2} P, \tag{2.28f}$$

$$\frac{\partial P_{z,3}}{\partial t} = \frac{\sigma}{\epsilon_0\epsilon_{r,\infty}} P_{z,1} - \frac{\sigma(\sigma_x + \sigma_y)}{\epsilon_0} E_z, \tag{2.28g}$$

$$\frac{\partial Q_x}{\partial t} = -\frac{\sigma_x}{\epsilon_0} Q_x + \frac{\mu_0\mu_r\sigma_x\left(\sigma_y - \sigma_x\right)}{\epsilon_0^2} H_x, \tag{2.28h}$$

$$\frac{\partial Q_y}{\partial t} = -\frac{\sigma_y}{\epsilon_0} Q_y + \frac{\mu_0\mu_r\sigma_y\left(\sigma_x - \sigma_y\right)}{\epsilon_0^2} H_y, \tag{2.28i}$$

在方程中标红的部分，与卢老师论文中不相同的部分，注意 $P_{z,2}, P_{z,3}$ 与原论文中定义不同。

将上述方程无量纲化后得到：

$$\epsilon_{r,\infty}\frac{\partial E_z}{\partial t} = \frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} - \left[\sigma + \epsilon_{r,\infty}(\sigma_x+\sigma_y)\right]E_z - \frac{\partial P}{\partial t} - (\sigma_x+\sigma_y)P + P_{z,1} + P_{z,2} + P_{z,3}, \tag{2.29a}$$

$$\mu_r\frac{\partial H_x}{\partial t} = -\frac{\partial E_z}{\partial y} - \mu_r(\sigma_y-\sigma_x)H_x + Q_x \tag{2.29b}$$

$$\mu_r\frac{\partial H_y}{\partial t} = \frac{\partial E_z}{\partial x} - \mu_r(\sigma_x-\sigma_y)H_y + Q_y \tag{2.29c}$$

$$P + \tau^\alpha\frac{\partial^\alpha P}{\partial t^\alpha} = (\epsilon_{r,s}-\epsilon_{r,\infty})E_z, \tag{2.29d}$$

$$\frac{\partial P_{z,1}}{\partial t} = -\epsilon_{r,\infty}\sigma_x\sigma_y E_z, \tag{2.29e}$$

total P_{z,1} w

$$\frac{\partial P_{z,2}}{\partial t} = -\sigma_x\sigma_y P, \tag{2.29f}$$

total P_{z,2} w

$$\frac{\partial P_{z,3}}{\partial t} = \frac{\sigma}{\epsilon_{r,\infty}}P_{z,1} - \sigma(\sigma_x+\sigma_y)E_z, \tag{2.29g}$$

total P_{z,3} w

$$\frac{\partial Q_x}{\partial t} = -\sigma_x Q_x + \mu_r\sigma_x\left(\sigma_y-\sigma_x\right)H_x, \tag{2.29h}$$

total Q_x witho

$$\frac{\partial Q_y}{\partial t} = -\sigma_y Q_y + \mu_r\sigma_y\left(\sigma_x-\sigma_y\right)H_y, \tag{2.29i}$$

total Q_y witho

3 Stability Analysis in 1D

Analysis in 1D

基于文章 [14] 所分析的在 Cole-Cole 介质中带有 Dirichlet 边界条件的 Maxwell 方程组的稳定性，将所研究的问题拓展到带有 UPML 吸收层情形。

首先给出频域下，带有 UPML 吸收边界层的 Cole-Cole 介质下的有关 E, H, P 的方程组，并假定$P \in H^1\left(0,T;L^2(\Omega)\right)$(这个假设可能过于强)：

$$\begin{aligned}
\frac{\partial \hat{H}}{\partial x} &= j\omega\left(\varepsilon_0\varepsilon_\infty\hat{E} + \hat{P}\right)s_z \\
\frac{\partial \hat{E}}{\partial x} &= j\omega\mu_0\hat{H}s_z \\
\hat{P} &= \varepsilon_0\left(\varepsilon_r-\varepsilon_\infty\right)\hat{E} = \varepsilon_0\frac{\varepsilon_s-\varepsilon_\infty}{1+(j\omega\tau)^\alpha}\hat{E}
\end{aligned} \tag{3.1}$$

Maxwell Equatio

其中，$s_z = 1 + \frac{\sigma_z}{j\omega\epsilon_0}$ 是有关 UPML 层的取值，具体的取值方法在 section 2中提及。

将方程组(3.1)转化为时域下的方程组：

$$\begin{aligned} \varepsilon_0\varepsilon_\infty\frac{\partial E}{\partial t} &= \frac{\partial H}{\partial x} - \frac{\partial P}{\partial t} - \sigma_z\varepsilon_\infty E - \frac{\sigma_z}{\varepsilon_0}P \\ \mu_0\frac{\partial H}{\partial t} &= \frac{\partial E}{\partial x} - \frac{\mu_0}{\varepsilon_0}\sigma_z H \\ P + \tau^\alpha\frac{\partial^\alpha P}{\partial t^\alpha} &= \varepsilon_0\left(\varepsilon_s - \varepsilon_\infty\right)E \end{aligned} \tag{3.2}$$

Maxwell Equatio

边界条件取做 perfectly conducting boundary condition:

$$\widehat{\boldsymbol{n}} \times E = 0 \quad \text{ on } \partial\Omega, \tag{3.3}$$

其中 $\hat{\boldsymbol{n}}$ 为 $\partial\Omega$ 上的外法向量，初值条件为：

$$E(x,0) = E_0(x), \quad H(x,0) = H_0(x), \quad P(x,0) = P_0(x) \tag{3.4}$$

将 $\frac{\partial^\alpha P}{\partial t^\alpha}$ 用辅助变量进行表示：

$$\frac{\partial^\alpha P(x,t)}{\partial t^\alpha} = \int_0^\infty \varphi(x,t;\lambda)d\lambda, \tag{3.5}$$

其中辅助变量 φ 由下述 ODE 方程进行描述

$$\begin{aligned} \frac{\partial\varphi}{\partial t} &= -\lambda\varphi + \frac{\sin(\pi\alpha)}{\pi}\lambda^{\alpha-1}\frac{\partial P}{\partial t}, \quad \lambda > 0, \\ \varphi(x,0;\lambda) &= 0, \quad \lambda > 0. \end{aligned} \tag{3.6}$$

Auxiliary Equat

也可以将上述 ODE 的解直接写做：

$$\varphi(x,t;\lambda) := \frac{\sin(\pi\alpha)}{\pi}\lambda^{\alpha-1}\int_0^t e^{-(t-\tau)\lambda}\frac{\partial P(x,\tau)}{\partial\tau}d\tau. \tag{3.7}$$

3.1 Energy Function

对方程组(3.2)和辅助方程(3.6)设计相应的能量函数，定义如下：

$$W = \int_\Omega\left(\frac{a_1}{2}\varepsilon_0\varepsilon_\infty E^2 + \frac{a_2}{2}\mu_0 H^2 + \frac{a_3}{2}P^2 + \frac{1}{2}\int_0^\infty a_4\varphi^2 d\lambda + \int_0^\infty b_1\varphi P d\lambda\right)dx, \tag{3.8}$$

1-D energy func

其中 $E, H, P, \varphi, \lambda$ 分别为电场强度、磁场强度、电极化强度、电极化强度的辅助变量，辅助变量的参数。并且其中 a_1, a_2 为标量，$a_3(x)$ 是关于空间变量 x 的函数,$a_4(x,\lambda), b_1(x,\lambda)$ 是关于空间变量 x 和 λ 的函数。

需要证明这个能量函数本身非负，且是随着时间衰减的，即可说明这个系统是稳定的。

将能量函数 W 对时间做导数：

$$\begin{aligned}\frac{dW}{dt} &= \int_\Omega \left[a_1\varepsilon_0\varepsilon_\infty E\frac{\partial E}{\partial t} + a_2\mu_0 H\frac{\partial H}{\partial t} + a_3 P\frac{\partial P}{\partial t} + \int_0^\infty a_4\varphi\frac{\partial \varphi}{\partial t}d\lambda + \int_0^\infty b_1\left(\frac{\partial \varphi}{\partial t}P + \frac{\partial P}{\partial t}\varphi\right)d\lambda \right] dx \\ &= \int_\Omega \left[a_1\frac{\partial H}{\partial x}E - a_1\frac{\partial P}{\partial t}E - a_1\sigma_z\varepsilon_\infty E^2 - a_1\frac{\sigma_z}{\varepsilon_0}PE + a_2\frac{\partial E}{\partial x}H - a_2\frac{\mu_0}{\varepsilon_0}\sigma_z H^2 + a_3 P\frac{\partial P}{\partial t} \right. \\ &\quad \left. + \int_0^\infty a_4\varphi\frac{\partial \varphi}{\partial t}d\lambda + \int_0^\infty b_1\varphi\frac{\partial P}{\partial t}d\lambda + \int_0^\infty b_1\left(-\lambda\varphi P + \frac{\sin\pi\alpha}{\pi}\lambda^{\alpha-1}\frac{\partial P}{\partial t}P\right)d\lambda \right] dx.\end{aligned} \tag{3.9}$$

再经过一系列化简，得到如下的表达：

$$\begin{aligned}\frac{dW}{dt} &= \int_\Omega \left[\left(a_1\frac{\partial H}{\partial x}E + a_2\frac{\partial E}{\partial x}H \right) - a_1\sigma_z\varepsilon_\infty E^2 - a_2\frac{\mu_0}{\varepsilon_0}\sigma_z H^2 - a_1\frac{\sigma_z}{\varepsilon_0}\frac{1}{\varepsilon_0(\varepsilon_s-\varepsilon_0)}P^2 \right. \\ &\quad - \int_0^\infty \left(a_1\frac{\sigma_z}{\varepsilon_0}\frac{\tau^\alpha}{\varepsilon_0(\varepsilon_s-\varepsilon_\infty)} + b_1\lambda \right)\varphi P d\lambda \\ &\quad + \left(-\frac{a_1}{\epsilon_0(\epsilon_s-\epsilon_\infty)} + a_3 + \int_0^\infty b_1\frac{\sin\pi\alpha}{\pi}\lambda^{\alpha-1}d\lambda \right)\epsilon_0(\epsilon_s-\epsilon_\infty)E\frac{\partial P}{\partial t} \\ &\quad + \int_0^\infty \left[a_4 + b_1\frac{\pi}{\sin\pi\alpha}\lambda^{1-\alpha} - \tau^\alpha\left(a_3 + \int_0^\infty b_1\frac{\sin\pi\alpha}{\pi}s^{\alpha-1}ds \right)\frac{\pi}{\sin\pi\alpha}\lambda^{1-\alpha} \right]\varphi\frac{\partial\varphi}{\partial t}d\lambda \\ &\quad \left. - \int_0^\infty \left[-\frac{\pi}{\sin\pi\alpha}\lambda^{2-\alpha}b_1 + \tau^\alpha\left(a_3 + \int_0^\infty b_1\frac{\sin\pi\alpha}{\pi}s^{\alpha-1}ds \right)\frac{\pi}{\sin\pi\alpha}\lambda^{2-\alpha} \right]\varphi^2 d\lambda \right] dx.\end{aligned} \tag{3.10}$$

dW/dt

对于空间导数项，由分部积分可知：

$$\int_\Omega E\frac{\partial H}{\partial x}dx = EH|_{\partial\Omega} - \int_\Omega H\frac{\partial E}{\partial x}dx,$$

UPML 吸收层的外层边界条件为 perfectly conducting boundary condition，对应于一维情形，则有：

$$\int_\Omega E\frac{\partial H}{\partial x} + H\frac{\partial E}{\partial x}dx = EH|_{\partial\Omega} = 0.$$

想要能量函数的导数是非正的，需要将 $\frac{\partial W}{\partial t}$ 凑出非正的形式。不妨对式(3.10)提出如下的假设

$$\begin{cases} \displaystyle\int_0^\infty b_1\frac{\sin\pi\alpha}{\pi}\lambda^{\alpha-1}d\lambda = \frac{a_1}{\epsilon_0(\epsilon_s-\epsilon_\infty)} - a_3 \\ \displaystyle a_4 = (\frac{\tau^\alpha}{\epsilon_0(\epsilon_s-\epsilon_\infty)}a_1 - b_1)\frac{\pi}{\sin\pi\alpha}\lambda^{1-\alpha} \\ a_1 = a_2 \end{cases} \tag{3.11}$$

这样式(3.10)中 $E\frac{\partial P}{\partial t}, \varphi\frac{\partial \varphi}{\partial t}$ 以及空间导数项对应的系数为 0。将 dW/dt 剩下的项配方形成负的平方项的形式。

定义 $M \triangleq \frac{\sigma_z}{\epsilon_0}\frac{1}{\epsilon_0(\epsilon_s-\epsilon_\infty)}$，其中 σ_z 为 UPML 层参数

$$\sigma_z = \sigma_{\max}\frac{|x-x_0|^m}{d^m} = -\frac{(m+1)ln(R(0))}{2\eta d}\frac{|x-x_0|^m}{d^m}$$

则式(3.10)在上述假设条件下化简为:

$$\begin{aligned}
\frac{dW}{dt} =& \int_\Omega \Bigg[-a_1\sigma_z\epsilon_\infty E^2 - a_2\frac{\mu_0}{\varepsilon_0}\sigma_z H^2 \\
& -a_1 M P^2 - \int_0^\infty (a_1\tau^\alpha M + b_1\lambda)\varphi P d\lambda - \int_0^\infty \lambda a_4\varphi^2 d\lambda\Bigg] dx \\
=& \int_\Omega \Bigg[-a_1\sigma_z\epsilon_\infty E^2 - a_2\frac{\mu_0}{\varepsilon_0}\sigma_z H^2 - (a_1 M - \int_0^\infty \frac{(a_1\tau^\alpha M + b_1\lambda)^2}{4\lambda a_4}d\lambda)P^2 \\
& - \int_0^\infty \frac{(a_1\tau^\alpha M + b_1\lambda)^2}{4\lambda a_4}P^2 d\lambda - \int_0^\infty (a_1\tau^\alpha M + b_1\lambda)\varphi P d\lambda - \int_0^\infty \lambda a_4\varphi^2 d\lambda\Bigg] dx \\
=& \int_\Omega \Bigg[-a_1\sigma_z\epsilon_\infty E^2 - a_2\frac{\mu_0}{\varepsilon_0}\sigma_z H^2 - \left(a_1 M - \int_0^\infty \frac{(a_1\tau^\alpha M + b_1\lambda)^2}{4\lambda a_4}d\lambda\right)P^2 \\
& - \int_0^\infty \left(\frac{(a_1\tau^\alpha M + b_1\lambda)}{2\sqrt{\lambda a_4}}P + \sqrt{\lambda a_4}\varphi\right)^2 d\lambda\Bigg] dx
\end{aligned} \tag{3.12}$$

故满足如下不等式时:

$$a_1 M - \int_0^\infty \frac{(a_1\tau^\alpha M + b_1\lambda)^2}{4\lambda a_4}d\lambda \geq 0, \tag{3.13}$$

inequality 3

能量函数的随时间递减，即 $dW/dt \leq 0$

同时需要验证能量函数 $W \geq 0$，即

$$\begin{aligned}
W =& \int_\Omega \left(\frac{a_1}{2}\varepsilon_0\varepsilon_\infty E^2 + \frac{a_2}{2}\mu_0 H^2 + \frac{a_3}{2}P^2 + \frac{1}{2}\int_0^\infty a_4\varphi^2 d\lambda + \int_0^\infty b_1\varphi P d\lambda\right) dx \\
=& \int_\Omega \left(\frac{a_1}{2}\varepsilon_0\varepsilon_\infty E^2 + \frac{a_2}{2}\mu_0 H^2 + \left(\frac{a_3}{2} - \frac{1}{2}\int_0^\infty \frac{b_1^2}{a_4}d\lambda\right)P^2\right. \\
& \left. + \frac{1}{2}\int_0^\infty \frac{b_1^2}{a_4}P^2 d\lambda + \frac{1}{2}\int_0^\infty a_4\varphi^2 d\lambda + \int_0^\infty b_1\varphi P d\lambda\right) dx \\
=& \int_\Omega \left(\frac{a_1}{2}\varepsilon_0\varepsilon_\infty E^2 + \frac{a_2}{2}\mu_0 H^2 + \left(\frac{a_3}{2} - \frac{1}{2}\int_0^\infty \frac{b_1^2}{a_4}d\lambda\right)P^2 + \frac{1}{2}\int_0^\infty \left(\frac{b_1}{\sqrt{a_4}}P + \sqrt{a_4}\varphi\right)^2 d\lambda\right) dx
\end{aligned} \tag{3.14}$$

故满足如下不等式时：

$$a_3 - \int_0^\infty \frac{b_1^2}{a_4} d\lambda \geq 0, \tag{3.15}$$

inequality 4

能量函数是正的，即 $W \geq 0$。将对能量函数所提的要求合并在一起：

$$\begin{cases} \int_0^\infty b_1 \dfrac{\sin \pi\alpha}{\pi} \lambda^{\alpha-1} d\lambda = \dfrac{a_1}{\epsilon_0(\epsilon_s - \epsilon_\infty)} - a_3 & \text{ineq:1} \quad (3.16a) \\ a_4 = (\dfrac{\tau^\alpha}{\epsilon_0(\epsilon_s - \epsilon_\infty)} a_1 - b_1) \dfrac{\pi}{\sin \pi\alpha} \lambda^{1-\alpha} & \text{ineq:2} \quad (3.16b) \\ a_1 = a_2 & \text{ineq:3} \quad (3.16c) \\ a_3 - \int_0^\infty \dfrac{b_1^2}{a_4} d\lambda \geq 0 & \text{ineq:4} \quad (3.16d) \\ a_1 M - \int_0^\infty \dfrac{(a_1 \tau^\alpha M + b_1 \lambda)^2}{4\lambda a_4} d\lambda \geq 0 & \text{ineq:5} \quad (3.16e) \end{cases}$$

定义 $C \triangleq \frac{\pi}{\sin \pi\alpha}$，对 $a_3(x), a_4(x,\lambda), b_1(x,\lambda)$ 取做如下的形式：

$$a_3(x) = s(x) a_1, \tag{3.17}$$

a_3

$$a_4(x,\lambda) = \begin{cases} \left(\dfrac{\tau^\alpha}{\epsilon_0(\epsilon_s - \epsilon_\infty)} + kM\lambda^\beta \right) C\lambda^{1-\alpha} a_1, & \lambda \in (0, \delta_0] \\ \left(\dfrac{\tau^\alpha}{\epsilon_0(\epsilon_s - \epsilon_\infty)} + \tau^\alpha M \lambda^{-1} \right) C\lambda^{1-\alpha} a_1, & \lambda \in (\delta_0, +\infty) \end{cases} \tag{3.18}$$

a_4

$$b_1(x,\lambda) = \begin{cases} -kM\lambda^\beta a_1, & \lambda \in (0, \delta_0] \\ -\tau^\alpha M \lambda^{-1} a_1, & \lambda \in (\delta_0, +\infty) \end{cases} \tag{3.19}$$

b_1

其中要求 $s(x) \geq 0, -\alpha < \beta < \alpha - 1$，且 k, β, δ_0 为可变参数。

上述 a_4, b_1 的取法是自然满足式(3.16b)的，将具体形式带入式(3.16a)中：

$$\begin{aligned} \int_0^{\delta_0} -kM\lambda^\beta \frac{1}{C} \lambda^{\alpha-1} d\lambda + \int_{\delta_0}^\infty -\tau^\alpha M \lambda^{-1} \frac{1}{C} \lambda^{\alpha-1} d\lambda &= \frac{1}{\epsilon_0(\epsilon_s - \epsilon_\infty)} - s \\ -\frac{kM}{C} \frac{\delta_0^{\alpha+\beta}}{\alpha+\beta} - \frac{\tau^\alpha M}{C} \frac{\delta_0^{\alpha-1}}{1-\alpha} &= \frac{1}{\epsilon_0(\epsilon_s - \epsilon_\infty)} - s \\ s &= \frac{1}{\epsilon_0(\epsilon_s - \epsilon_\infty)} + \frac{kM}{C} \frac{\delta_0^{\alpha+\beta}}{\alpha+\beta} + \frac{\tau^\alpha M}{C} \frac{\delta_0^{\alpha-1}}{1-\alpha} \end{aligned} \tag{3.20}$$

在 $\alpha < 1$ 时，满足 $s > 0$.

验证不等式(3.16d)，带入 a_4, b_1 表达式：

$$\begin{aligned}\int_0^\infty \frac{b_1^2}{a_4}d\lambda &= \int_0^{\delta_0} \frac{k^2M^2\lambda^{2\beta}}{(\frac{\tau^\alpha}{\epsilon_0(\epsilon_s-\epsilon_\infty)}+kM\lambda^\beta)C\lambda^{1-\alpha}}a_1 d\lambda + \int_{\delta_0}^\infty \frac{\tau^{2\alpha}M^2\lambda^{-2}}{\left(\frac{\tau^\alpha}{\epsilon_0(\epsilon_s-\epsilon_\infty)}+\tau^\alpha M\lambda^{-1}\right)C\lambda^{1-\alpha}}a_1 d\lambda \\ &\le \int_0^{\delta_0}\frac{kM}{C}\lambda^{\alpha+\beta-1}a_1 d\lambda + \int_{\delta_0}^\infty \frac{\tau^\alpha M}{C}\lambda^{\alpha-2}a_1 d\lambda \\ &= \frac{kM}{C}\frac{\delta_0^{\alpha+\beta}}{\alpha+\beta}a_1 + \frac{\tau^\alpha M}{C}\frac{\delta_0^{\alpha-1}}{1-\alpha}a_1 \\ &\le sa_1 = a_3\end{aligned} \tag{3.21}$$

验证不等式(3.16e)，同样带入 a_4, b_1 表达式：

$$\begin{aligned}&\int_0^\infty \frac{(a_1\tau^\alpha M + b_1\lambda)^2}{4\lambda a_4}d\lambda \\ =&\int_0^{\delta_0}\frac{(a_1\tau^\alpha M - kM\lambda^{\beta+1}a_1)^2}{4(\frac{\tau^\alpha}{\epsilon_0(\epsilon_s-\epsilon_\infty)}+kM\lambda^\beta)C\lambda^{2-\alpha}a_1}d\lambda + \int_{\delta_0}^\infty \frac{(\tau^\alpha M a_1 - \tau^\alpha M\lambda^{-1}\lambda a_1)^2}{4(\frac{\tau^\alpha}{\epsilon_0(\epsilon_s-\epsilon_\infty)}+\tau^\alpha M\lambda^{-1})C\lambda^{2-\alpha}a_1}d\lambda \\ =&\int_0^{\delta_0}\frac{(a_1\tau^\alpha M - kM\lambda^{\beta+1}a_1)^2}{4(\frac{\tau^\alpha}{\epsilon_0(\epsilon_s-\epsilon_\infty)}+kM\lambda^\beta)C\lambda^{2-\alpha}a_1}d\lambda \\ \le&\int_0^{\delta_0}\frac{\tau^{2\alpha}M^2 - 2\tau^\alpha kM^2\lambda^{\beta+1}+k^2M^2\lambda^{2\beta+2}}{4kMC\lambda^{\beta+2-\alpha}}a_1 d\lambda \\ =&\int_0^{\delta_0}\left(\frac{\tau^{2\alpha}M}{4kC}\lambda^{\alpha-\beta-2} - \frac{\tau^\alpha M}{2C}\lambda^{\alpha-1} + \frac{kM}{4C}\lambda^{\alpha+\beta}\right)a_1 d\lambda \\ =&\left(\frac{\tau^{2\alpha}M}{4kC}\frac{\delta_0^{\alpha-\beta-1}}{\alpha-\beta-1} - \frac{\tau^\alpha M}{2C}\frac{\delta_0^\alpha}{\alpha} + \frac{kM}{4C}\frac{\delta_0^{\alpha+\beta+1}}{\alpha+\beta+1}\right)a_1\end{aligned} \tag{3.22}$$

则只需证：

$$\begin{aligned}\left(\frac{\tau^{2\alpha}M}{4kC}\frac{\delta_0^{\alpha-\beta-1}}{\alpha-\beta-1} - \frac{\tau^\alpha M}{2C}\frac{\delta_0^\alpha}{\alpha} + \frac{kM}{4C}\frac{\delta_0^{\alpha+\beta+1}}{\alpha+\beta+1}\right)a_1 &\le a_1 M \\ \frac{\tau^{2\alpha}}{4kC}\frac{\delta_0^{\alpha-\beta-1}}{\alpha-\beta-1} + \frac{k}{4C}\frac{\delta_0^{\alpha+\beta+1}}{\alpha+\beta+1} - \frac{\tau^\alpha}{2C}\frac{\delta_0^\alpha}{\alpha} - 1 &\le 0\end{aligned} \tag{3.23}$$

当 $k = \tau^\alpha\sqrt{\frac{\alpha+\beta+1}{\alpha-\beta-1}}\delta_0^{-\beta-1}$ 时，

$$\frac{\tau^\alpha\delta_0^\alpha}{2C}\left(\sqrt{\frac{1}{(\alpha-\beta-1)(\alpha+\beta+1)}} - \frac{1}{\alpha}\right) \le 1 \tag{3.24}$$

则

$$\delta_0 \le \frac{(2C)^{\frac{1}{\alpha}}}{\tau}\left(\frac{1}{\sqrt{\frac{1}{(\alpha-\beta-1)(\alpha+\beta+1)}}-\frac{1}{\alpha}}\right)^{\frac{1}{\alpha}} \tag{3.25}$$

delta_0 inequal

总结下来，各个参数的取法为

$$\begin{cases} a_1 = a_2 = 1 \\ a_3 = \dfrac{1}{\epsilon_0(\epsilon_s-\epsilon_\infty)} + \dfrac{\tau^\alpha\sigma_z}{C\epsilon_0^2(\epsilon_s-\epsilon_\infty)}\sqrt{\dfrac{\alpha+\beta+1}{\alpha-\beta-1}}\dfrac{\delta_0^{\alpha-1}}{\alpha+\beta} + \dfrac{\tau^\alpha\sigma_z}{C\epsilon_0^2(\epsilon_s-\epsilon_\infty)}\dfrac{\delta_0^{\alpha-1}}{1-\alpha} \\ a_4 = \begin{cases} \left(\dfrac{\tau^\alpha}{\epsilon_0(\epsilon_s-\epsilon_\infty)} + \dfrac{\tau^\alpha\sigma_z}{\epsilon_0^2(\epsilon_s-\epsilon_\infty)}\sqrt{\dfrac{\alpha+\beta+1}{\alpha-\beta-1}}\delta_0^{-\beta-1}\lambda^\beta\right)C\lambda^{1-\alpha}a_1, & \lambda\in(0,\delta_0] \\ \left(\dfrac{\tau^\alpha}{\epsilon_0(\epsilon_s-\epsilon_\infty)} + \tau^\alpha\dfrac{\sigma_z}{\epsilon_0^2(\epsilon_s-\epsilon_\infty)}\lambda^{-1}\right)C\lambda^{1-\alpha}a_1, & \lambda\in(\delta_0,+\infty) \end{cases} \\ b_1 = \begin{cases} -\dfrac{\tau^\alpha\sigma_z}{\epsilon_0^2(\epsilon_s-\epsilon_\infty)}\sqrt{\dfrac{\alpha+\beta+1}{\alpha-\beta-1}}\delta_0^{-\beta-1}\lambda^\beta a_1, & \lambda\in(0,\delta_0] \\ -\tau^\alpha\dfrac{\sigma_z}{\epsilon_0^2(\epsilon_s-\epsilon_\infty)}\lambda^{-1}a_1, & \lambda\in(\delta_0,+\infty) \end{cases} \end{cases} \tag{3.26}$$

其中 $0.5<\alpha<1, -\alpha<\beta<\alpha-1$，$\delta_0$ 的取法由式(3.25)给出。

当 $\sigma_z=0$ 时，即解算的 Maxwell 方程组对应于计算色散媒质中的 Maxwell 方程组，能量函数 W 也退化为 [14] 中所提出的形式。

回到能量函数定义式，还需要说明按照所取的 $a_1,a_2,a_3(x),a_4(x,\lambda),b_1(x,\lambda)$ 能量函数(3.8)是 well-defined .

带入 $a_4(x,\lambda),\varphi$ 的定义，对于能量函数的 λ 积分项 $\int_0^\infty a_4\varphi^2 d\lambda$，需要验证其是否收敛。

$$\begin{aligned} \int_0^\infty a_4\varphi^2 d\lambda =& \int_0^{\delta_0}\left(\frac{\tau^\alpha}{\epsilon_0(\epsilon_s-\epsilon_\infty)}+kM\lambda^\beta\right)C\lambda^{1-\alpha}a_1\varphi^2 d\lambda \\ &+\int_{\delta_0}^\infty\left(\frac{\tau^\alpha}{\epsilon_0(\epsilon_s-\epsilon_\infty)}+\tau^\alpha M\lambda^{-1}\right)C\lambda^{1-\alpha}a_1\varphi^2 d\lambda \\ =&\frac{\tau^\alpha}{\epsilon_0(\epsilon_s-\epsilon_\infty)}Ca_1\int_0^{\delta_0}\lambda^{1-\alpha}\varphi^2 d\lambda + kMCa_1\int_0^{\delta_0}\lambda^{\beta-\alpha+1}\varphi^2 d\lambda \\ &+\frac{\tau^\alpha}{\epsilon_0(\epsilon_s-\epsilon_\infty)}Ca_1\int_{\delta_0}^\infty\lambda^{1-\alpha}\varphi^2 d\lambda + \tau^\alpha MCa_1\int_{\delta_0}^\infty\lambda^{-\alpha}\varphi^2 d\lambda \end{aligned} \tag{3.27}$$

由于 $P \in H^1\left(0,T;L^2(\Omega)\right)$，则由 Cauchy-Schwartz 不等式：

$$\begin{aligned}\left(\int_0^t e^{-(t-\tau)\lambda}\frac{\partial P(x,\tau)}{\partial \tau}d\tau\right)^2 &\le \left(\int_0^t \left|e^{-(t-\tau)\lambda}\right|^2 d\tau\right)\left(\int_0^t \left|\frac{\partial P(x,\tau)}{\partial \tau}\right|^2 d\tau\right)\\ &= \frac{1-e^{-2t\lambda}}{2\lambda}\int_0^t \left|\frac{\partial P(x,\tau)}{\partial \tau}\right|^2 d\tau\end{aligned} \tag{3.28}$$

由

$$\frac{1-e^{-2t\lambda}}{2\lambda} \le t$$

对于第一项，带入 $\varphi(x,t;\lambda) := \frac{1}{C}\lambda^{\alpha-1}\int_0^t e^{-(t-\tau)\lambda}\frac{\partial P(x,\tau)}{\partial \tau}d\tau$

$$\begin{aligned}\int_0^{\delta_0}\lambda^{1-\alpha}\varphi^2 d\lambda &= \frac{1}{C^2}\int_0^{\delta_0}\lambda^{\alpha-1}\left(\int_0^t e^{-(t-\tau)\lambda}\frac{\partial P(x,\tau)}{\partial \tau}d\tau\right)^2 d\lambda\\ &\le \frac{1}{C^2}\int_0^{\delta_0}\lambda^{\alpha-1}t\int_0^t \left|\frac{\partial P(x,\tau)}{\partial \tau}\right|^2 d\tau d\lambda\\ &= \frac{1}{C^2}\frac{\delta_0^\alpha}{\alpha}t\int_0^t \left|\frac{\partial P(x,\tau)}{\partial \tau}\right|^2 d\tau\end{aligned} \tag{3.29}$$

由 α,δ_0 的取值范围可知其是收敛的，同理对于第二项积分

$$\begin{aligned}\int_0^{\delta_0}\lambda^{\beta-\alpha+1}\varphi^2 d\lambda &= \frac{1}{C^2}\int_0^{\delta_0}\lambda^{\beta-\alpha+1}\left(\int_0^t e^{-(t-\tau)\lambda}\frac{\partial P(x,\tau)}{\partial \tau}d\tau\right)^2 d\lambda\\ &\le \frac{1}{C^2}\int_0^{\delta_0}\lambda^{\beta-\alpha+1}t\int_0^t \left|\frac{\partial P(x,\tau)}{\partial \tau}\right|^2 d\tau d\lambda\\ &= \frac{1}{C^2}\frac{\delta_0^{\beta-\alpha+2}}{\beta-\alpha+2}t\int_0^t \left|\frac{\partial P(x,\tau)}{\partial \tau}\right|^2 d\tau\end{aligned} \tag{3.30}$$

由 α,β,δ_0 的取值范围可知其是收敛的，对于第三项积分

$$\begin{aligned}\int_{\delta_0}^{\infty}\lambda^{1-\alpha}\psi^2 d\lambda &= \frac{1}{C^2}\int_{\delta_0}^{\infty}\lambda^{\alpha-1}\left(\int_0^t e^{-(t-\tau)\lambda}\frac{\partial P(x,\tau)}{\partial \tau}d\tau\right)^2 d\lambda\\ &\le \frac{1}{C^2}\int_{\delta_0}^{\infty}\lambda^{\alpha-1}\frac{1-e^{-2t\lambda}}{2\lambda}\int_0^t \left|\frac{\partial P(x,\tau)}{\partial \tau}\right|^2 d\tau d\lambda\\ &\le \frac{1}{2C^2}\int_{\delta_0}^{\infty}\lambda^{\alpha-2}\int_0^t \left|\frac{\partial P(x,\tau)}{\partial \tau}\right|^2 d\tau d\lambda\\ &= \frac{1}{2C^2}\frac{\delta_0^{\alpha-1}}{1-\alpha}\int_0^t \left|\frac{\partial P(x,\tau)}{\partial \tau}\right|^2 d\tau\end{aligned} \tag{3.31}$$

由 α 的取值范围可知其是收敛的，对于第四项积分

$$\begin{aligned}\int_{\delta_0}^{\infty}\lambda^{-\alpha}\varphi^2 d\lambda &= \frac{1}{C^2}\int_{\delta_0}^{\infty}\lambda^{\alpha-2}\left(\int_0^t e^{-(t-\tau)\lambda}\frac{\partial P(x,\tau)}{\partial\tau}d\tau\right)^2 d\lambda\\ &\le \frac{1}{C^2}\int_{\delta_0}^{\infty}\lambda^{\alpha-2}\frac{1-e^{-2t\lambda}}{2\lambda}\int_0^t\left|\frac{\partial P(x,\tau)}{\partial\tau}\right|^2 d\tau d\lambda\\ &\le \frac{1}{2C^2}\int_{\delta_0}^{\infty}\lambda^{\alpha-3}\int_0^t\left|\frac{\partial P(x,\tau)}{\partial\tau}\right|^2 d\tau d\lambda\\ &= \frac{1}{2C^2}\frac{\delta_0^{\alpha-2}}{2-\alpha}\int_0^t\left|\frac{\partial P(x,\tau)}{\partial\tau}\right|^2 d\tau\end{aligned} \tag{3.32}$$

由 α 的取值范围可知其是收敛的。故 $\int_{)}^{\infty} a_4\varphi^2 d\lambda$ 是收敛的。

对于 $\int_0^{\infty} b_1\varphi d\lambda$，同样需要验证其是否收敛

$$\int_0^{\infty} b_1\varphi d\lambda = -\int_0^{\delta_0} kM\lambda^{\beta}a_1\varphi d\lambda - \int_{\delta_0}^{\infty}\tau^{\alpha}M\lambda^{-1}a_1\varphi d\lambda \tag{3.33}$$

对于第一项积分

$$\begin{aligned}\int_0^{\delta_0}\lambda^{\beta}\varphi d\lambda &= \frac{1}{C}\int_0^{\delta_0}\lambda^{\beta+\alpha-1}\int_0^t e^{-(t-\tau)\lambda}\frac{\partial P(x,\tau)}{\partial\tau}d\tau d\lambda\\ -\frac{1}{C}\int_0^{\delta_0}\lambda^{\beta+\alpha-1}\int_0^t\left|\frac{\partial P(x,\tau)}{\partial\tau}\right|d\tau d\lambda &\le \int_0^{\delta_0}\lambda^{\beta}\varphi d\lambda \le \frac{1}{C}\int_0^{\delta_0}\lambda^{\beta+\alpha-1}\int_0^t\left|\frac{\partial P(x,\tau)}{\partial\tau}\right|d\tau d\lambda\\ -\frac{1}{C}\frac{\delta_0^{\alpha+\beta}}{\alpha+\beta}\int_0^t\left|\frac{\partial P(x,\tau)}{\partial\tau}\right|d\tau &\le \int_0^{\delta_0}\lambda^{\beta}\varphi d\lambda \le \frac{1}{C}\frac{\delta_0^{\alpha+\beta}}{\alpha+\beta}\int_0^t\left|\frac{\partial P(x,\tau)}{\partial\tau}\right|d\tau\end{aligned} \tag{3.34}$$

而

$$\left(\int_0^t\left|\frac{\partial P(x,\tau)}{\partial\tau}\right|d\tau\right)^2 \le \left(\int_0^t\left|\frac{\partial P(x,\tau)}{\partial\tau}\right|^2 d\tau\right)\left(\int_0^t 1 d\tau\right)$$

则由 α,β 取值范围和 $P\in H^1\left(0,T;L^2(\Omega)\right)$ 可知第一项积分是收敛的，同理对于第二项积分

$$\begin{aligned}\int_{\delta_0}^{\infty}\lambda^{-1}\varphi d\lambda &= \frac{1}{C}\int_{\delta_0}^{\infty}\lambda^{\alpha-2}\int_0^t e^{-(t-\tau)\lambda}\frac{\partial P(x,\tau)}{\partial\tau}d\tau d\lambda\\ -\frac{1}{C}\int_{\delta_0}^{\infty}\lambda^{\alpha-2}\int_0^t\left|\frac{\partial P(x,\tau)}{\partial\tau}\right|d\tau d\lambda &\le \int_{\delta_0}^{\infty}\lambda^{-1}\varphi d\lambda \le \frac{1}{C}\int_{\delta_0}^{\infty}\lambda^{\alpha-2}\int_0^t\left|\frac{\partial P(x,\tau)}{\partial\tau}\right|d\tau d\lambda\\ -\frac{1}{C}\frac{\delta_0^{\alpha-1}}{1-\alpha}\int_0^t\left|\frac{\partial P(x,\tau)}{\partial\tau}\right|d\tau &\le \int_{\delta_0}^{\infty}\lambda^{-1}\varphi d\lambda \le \frac{1}{C}\frac{\delta_0^{\alpha-1}}{1-\alpha}\int_0^t\left|\frac{\partial P(x,\tau)}{\partial\tau}\right|d\tau\end{aligned} \tag{3.35}$$

则由 α 的取值范围可知其是收敛的。故 $\int_0^\infty b_1\varphi d\lambda$ 是收敛的。

至此，我们证明完成了能量函数 $W \geq 0$ 是良定的，并且 $dW/dt \leq 0$ 恒成立，故对带有 UPML 吸收层的一维 Cole-Cole 介质中的 Maxwell 方程组是稳定的。

并且我们的一个特点是，吸收参数 σ 不是常数。参见 [2]

3.2 Semi-discrete DG Scheme

We divide the interval $\Omega = [x_-, x_+]$ into M subintervals:$x_- = x_{\frac{1}{2}} < x_{\frac{3}{2}} < \cdots < x_{M+\frac{1}{2}} = x_+$, and the mesh size is denoted by $h_j = x_{j+\frac{1}{2}} - x_{j-\frac{1}{2}}$ with $h = \max_j h_j, \rho = \min_j h_j$. We assume that the mesh is regular, namely, $\exists\gamma, \gamma h \leq \rho \leq h$. The j-th cell is denoted by $I_j = [x_{j-\frac{1}{2}}, x_{j+\frac{1}{2}}]$, and the center of each cell is $x_j = (x_{j-\frac{1}{2}} + x_{j+\frac{1}{2}})/2$. The piecewise polynomial space is defined as

$$V_h = \left\{v : v|_{I_j} \in \mathbb{P}^k, \quad j = 1, \cdots, M\right\}, k \in \mathbb{N}, k \geq 1, \tag{3.36}$$

where $\mathbb{P}^k$ is the space of polynomials of degree up to k in each cell I_j. For any function $v \in V_h$, we denote by $v^+_{j+\frac{1}{2}}$ and $v^-_{j+\frac{1}{2}}$ the limit values of v at $x_{j+1/2}$ from the right cell I_{j+1} and the left cell I_j, respectively. The jump and average at $x_{j+1/2}$ are defined as

$$[\![v]\!]_{j+\frac{1}{2}} = v^+_{j+\frac{1}{2}} - v^-_{j+\frac{1}{2}}, \quad \{\{v\}\}_{j+\frac{1}{2}} = \frac{1}{2}\left(v^+_{j+\frac{1}{2}} + v^-_{j+\frac{1}{2}}\right).$$

For the average and jump functions, the following identity holds:

$$\{\{a\}\}[\![b]\!] + \{\{b\}\}[\![a]\!] = [\![ab]\!]. \tag{3.37}$$

eq:jump average

The DG scheme for (3.2) is to find for any time $t \in (0, T]$ the unique $U_h(\cdot, t) \in V_h$

with $U = E, H, P, \varphi$ such that for all $v \in V_h$, the following holds:

$$\int_{I_j} \mu_0 \frac{\partial H_h}{\partial t} v dx = \widehat{E_h} v^-|_{x_{j+1/2}} - \widehat{E_h} v^+|_{x_{j-1/2}} - \int_{I_j} E_h \frac{\partial v}{\partial x} dx - \int_{I_j} \frac{\mu_0}{\epsilon_0} \sigma_x H_h v dx, \tag{3.38a}$$

semi H

$$\int_{I_j} \epsilon_0 \epsilon_{r,\infty} \frac{\partial E_h}{\partial t} v dx = \widehat{H_h} v^-|_{x_{j+1/2}} - \widehat{H_h} v^+|_{x_{j-1/2}} - \int_{I_j} H_h \frac{\partial v}{\partial x} dx - \int_{I_j} \frac{\partial P_h}{\partial t} v dx - \int_{I_j} \sigma_x \epsilon_{r,\infty} E_h v dx - \int_{I_j} \frac{\sigma_x}{\epsilon_0} P_h v dx, \tag{3.38b}$$

semi E

$$\int_{I_j} P_h v dx + \int_{I_j} \int_0^\infty \tau^\alpha \varphi_h v d\lambda dx = \int_{I_j} \epsilon_0 \left(\epsilon_{r,s} - \epsilon_{r,\infty}\right) E_h v dx, \tag{3.38c}$$

semi P

$$\int_{I_j} \frac{\partial \varphi_h}{\partial t} v dx = -\lambda \int_{I_j} \varphi_h v dx + \frac{\sin(\pi\alpha)}{\pi} \lambda^{\alpha-1} \int_{I_j} \frac{\partial P_h}{\partial t} v dx, \tag{3.38d}$$

semi varphi

where $\hat{\cdot}$ indicates the cell boundary terms obtained deom integration by parts, and they are so called numerical fluxes. The numerical fluxes are single-valued functions defined on the cell boundaries. We use the up wind numerical fluxes on the interior faces, which are given by

$$\widehat{E_h} = \{\{E_h\}\} + \frac{1}{2}\sqrt{\frac{\mu_0}{\epsilon_0 \epsilon_{r,\infty}}} [\![H_h]\!], \quad \widehat{H_h} = \{\{H_h\}\} + \frac{1}{2}\sqrt{\frac{\epsilon_0 \epsilon_{r,\infty}}{\mu_0}} [\![E_h]\!], \tag{3.39}$$

eq:upwind flux

and we choose the fluxes on the boundaries of Ω, $E_{h_{1/2}}^- = -E_{h_{1/2}}^+, E_{h_{M+1/2}}^+ = -E_{h_{M+1/2}}^-, H_{h_{1/2}}^- = H_{h_{1/2}}^+, H_{h_{M+1/2}}^+ = H_{h_{M+1/2}}^-$, and taking them into (3.39)

$$\begin{aligned} \widehat{E_h}_{1/2} &= \widehat{E_h}_{M+1/2} = 0, \\ \widehat{H_h}_{1/2} &= H_{h_{1/2}}^+ + \sqrt{\frac{\epsilon_0 \epsilon_{r,\infty}}{\mu_0}} E_{h_{1/2}}^+, \\ \widehat{H_h}_{M+1/2} &= H_{h_{M+1/2}}^- - \sqrt{\frac{\epsilon_0 \epsilon_{r,\infty}}{\mu_0}} E_{h_{M+1/2}}^-. \end{aligned} \tag{3.40}$$

也可以采用别的数值通量，如 20250412.md 中所提的

Now we can show the energy decreasing property of the semi-discrete DG scheme.

Theorem 3.1. *Under the PEC boundary condition, the initial condition, and the upwind numerical flux, the solution of semi-discrete DG scheme statisfies the following stability*

$$\frac{d}{dt} \mathscr{E}(E_h(t), H_h(t), P_h(t), \varphi_h(t)) \leq 0, \tag{3.41}$$

ergy decreasing

where $\mathscr{E}$ is given above.

证明. Setting $v = H_h$ in (3.38a) and $v = E_h$ in (3.38b), and summing up over all elements, because of $a_1 = a_2$, we have

$$\begin{aligned}
&\int_\Omega a_1\epsilon_0\epsilon_{r,\infty}\frac{\partial E_h}{\partial t}E_h dx + \int_\Omega a_2\mu_0\frac{\partial H_h}{\partial t}H_h dx \\
=&-\sum_{j=1}^{M}\int_{I_j} a_1\partial_x\left(E_hH_h\right)dx \\
&+\sum_{j=1}^{M}\left(a_1\widehat{E_h}H_h^-|_{j+1/2} - a_1\widehat{E_h}H_h^+|_{j-1/2} + a_1\widehat{H_h}E_h^-|_{j+1/2} - a_1\widehat{H_h}E_h^+|_{j-1/2}\right) \\
&-\int_\Omega a_1\frac{\mu_0}{\epsilon_0}\sigma_x H_h^2 dx - \int_\Omega a_1\sigma_x\epsilon_{r,\infty}E_h^2 dx - \int_\Omega a_1\frac{\sigma_x}{\epsilon_0}P_hE_h dx - \int_\Omega a_1\frac{\partial P_h}{\partial t}E_h dx,
\end{aligned} \tag{3.42}$$

eq:semi H E

where

$$\begin{aligned}
\sum_{j=1}^{M}\widehat{E_h}H_h^-|_{j+1/2} - \widehat{E_h}H_h^+|_{j-1/2} &= -\sum_{j=1}^{M-1}\widehat{E_h}\left[\!\left[H_h\right]\!\right]|_{j+1/2} - \widehat{E_h}_{1/2}H_h{}^+_{1/2} + \widehat{E_h}_{M+1/2}H_h{}^-_{M+1/2}, \\
\sum_{j=1}^{M}\widehat{H_h}E_h^-|_{j+1/2} - \widehat{H_h}E_h^+|_{j-1/2} &= -\sum_{j=1}^{M-1}\widehat{H_h}\left[\!\left[E_h\right]\!\right]|_{j+1/2} - \widehat{H_h}_{1/2}E_h{}^+_{1/2} + \widehat{H_h}_{M+1/2}E_h{}^-_{M+1/2}.
\end{aligned} \tag{3.43}$$

According to (3.37), we have

$$\left(\widehat{E_h}[\![H_h]\!] + \widehat{H_h}[\![E_h]\!]\right)|_{j+1/2} = [\![E_hH_h]\!]|_{j+1/2} + \mathscr{M}_{j+1/2}(E_h, H_h), \tag{3.44}$$

where

$$\mathscr{M}_{j+1/2}(E_h, H_h) = \frac{1}{2}\left(\sqrt{\frac{\mu_0}{\epsilon_0\epsilon_{r,\infty}}}[\![H_h]\!]^2 + \sqrt{\frac{\epsilon_0\epsilon_{r,\infty}}{\mu_0}}[\![E_h]\!]\right)^2_{j+1/2}. \tag{3.45}$$

Meanwhile

$$\begin{aligned}
\sum_{j=1}^{M}\int_{I_j}\partial_x\left(E_hH_h\right)dx &= \sum_{j=1}^{M}\left(E_hH_h\right)|_{j+1/2}^- - \left(E_hH_h\right)|_{j-1/2}^+ \\
&= \left(E_hH_h\right)|_{M+1/2}^- - \left(E_hH_h\right)|_{1/2}^+ - \sum_{j=1}^{M-1}[\![E_hH_h]\!]|_{j+1/2}.
\end{aligned} \tag{3.46}$$

As a result, the spatial part of the semi-discrete energy function i.e. the first two lines of (3.42) is reduced to

$$\begin{aligned}\mathscr{E}_s = &-\sum_{j=1}^{M-1} \mathscr{M}_{j+1/2}(E_h, H_h) - \widehat{E_h}_{1/2} H_{h\,1/2}^{+} - \widehat{H_h}_{1/2} E_{h\,1/2}^{+} + (E_h H_h)_{1/2}^{+} \\ &+ \widehat{E_h}_{M+1/2} H_{h\,M+1/2}^{-} + \widehat{H_h}_{M+1/2} E_{h\,M+1/2}^{-} - (E_h H_h)_{M+1/2}^{-}\,,\end{aligned} \tag{3.47}$$

taking the fluxes we choose above into the equation, we have

$$\mathscr{E}_s = -\sum_{j=1}^{M-1} \mathscr{M}_{j+1/2}(E_h, H_h) - \sqrt{\frac{\epsilon_0 \epsilon_{r,\infty}}{\mu_0}} \left(E_{h\,1/2}^{+}\right)^2 - \sqrt{\frac{\epsilon_0 \epsilon_{r,\infty}}{\mu_0}} \left(E_{h\,M+1/2}^{-}\right)^2 \leq 0. \tag{3.48}$$

Now for equations (3.38c), (3.38d), we do the same in the continuous case, the time derivative of the energy function. We recall the definition of the energy function in semi-discrete case

$$\mathscr{E}_h = \int_\Omega \left(\frac{a_1}{2} \epsilon_0 \epsilon_{r,\infty} E_h^2 + \frac{a_2}{2} \mu_0 H_h^2 + \frac{a_3}{2} P_h^2 + \frac{1}{2} \int_0^\infty a_4 \varphi_h^2 d\lambda + \int_0^\infty b_1 \varphi_h P_h d\lambda \right) dx, \tag{3.49}$$

eq: definition

In order to keep the positiveness of the energy, it's obviously that

$$\begin{aligned}\mathscr{E}_h = \int_\Omega \Bigg(&\frac{a_1}{2} \varepsilon_0 \varepsilon_\infty E_h^2 + \frac{a_2}{2} \mu_0 H_h^2 + \left(\frac{a_3}{2} - \frac{1}{2} \int_0^\infty \frac{b_1^2}{a_4} d\lambda \right) P_h^2 \\ &+ \frac{1}{2} \int_0^\infty \left(\frac{b_1}{\sqrt{a_4}} P_h + \sqrt{a_4} \varphi_h \right)^2 d\lambda \Bigg) dx,\end{aligned} \tag{3.50}$$

so $\mathscr{E}_h \geq 0$ is equivalent to

$$a_3 - \int_0^\infty \frac{b_1^2}{a_4} d\lambda \geq 0. \tag{3.51}$$

so

$$
\begin{aligned}
\frac{d\mathscr{E}_h}{dt} =& \int_\Omega \left[a_1\epsilon_0\epsilon_{r,\infty} E_h \frac{\partial E_h}{\partial t} + a_2\mu_0 H_h \frac{\partial H_h}{\partial t} + a_3 P_h \frac{\partial P_h}{\partial t} + \int_0^\infty a_4 \varphi_h \frac{\partial \varphi_h}{\partial t} d\lambda + \int_0^\infty b_1 \left(\frac{\partial \varphi_h}{\partial t} P_h + \frac{\partial P_h}{\partial t} \varphi_h \right) d\lambda \right] d\\
=& \mathscr{E}_s + \int_\Omega \left[-a_1 \frac{\mu_0}{\epsilon_0} \sigma_x H_h^2 - a_1 \sigma_x \epsilon_{r,\infty} E_h^2 - a_1 \frac{\sigma_x}{\epsilon_0} P_h E_h - a_1 \frac{\partial P_h}{\partial t} E_h + a_3 P_h \frac{\partial P_h}{\partial t} \right.\\
&+ \int_0^\infty a_4 \varphi_h \frac{\partial \varphi_h}{\partial t} d\lambda + \int_0^\infty b_1 \varphi_h \frac{\partial P_h}{\partial t} d\lambda + \left. \int_0^\infty b_1 \left(-\lambda \varphi_h P_h + \frac{\sin \pi\alpha}{\pi} \lambda^{\alpha-1} \frac{\partial P_h}{\partial t} P_h \right) d\lambda \right] dx\\
=& \mathscr{E}_s + \int_\Omega \left[-a_1 \sigma_x \epsilon_{r,\infty} E_h^2 - a_2 \frac{\mu_0}{\epsilon_0} \sigma_x H_h^2 - a_1 \frac{\sigma_x}{\epsilon_0} \frac{1}{\epsilon_0 (\epsilon_{r,s} - \epsilon_0)} P_h^2 \right.\\
&- \int_0^\infty \left(a_1 \frac{\sigma_x}{\epsilon_0} \frac{\tau^\alpha}{\epsilon_0 (\epsilon_{r,s} - \epsilon_{r,\infty})} + b_1 \lambda \right) \varphi_h P_h d\lambda\\
&+ \left(-\frac{a_1}{\epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty})} + a_3 + \int_0^\infty b_1 \frac{\sin \pi\alpha}{\pi} \lambda^{\alpha-1} d\lambda \right) \epsilon_0 (\epsilon_{r,s} - \epsilon_{r,\infty}) E_h \frac{\partial P_h}{\partial t}\\
&+ \int_0^\infty \left[a_4 + b_1 \frac{\pi}{\sin \pi\alpha} \lambda^{1-\alpha} - \tau^\alpha \left(a_3 + \int_0^\infty b_1 \frac{\sin \pi\alpha}{\pi} s^{\alpha-1} ds \right) \frac{\pi}{\sin \pi\alpha} \lambda^{1-\alpha} \right] \varphi_h \frac{\partial \varphi_h}{\partial t} d\lambda\\
&- \left. \int_0^\infty \left[-\frac{\pi}{\sin \pi\alpha} \lambda^{2-\alpha} b_1 + \tau^\alpha \left(a_3 + \int_0^\infty b_1 \frac{\sin \pi\alpha}{\pi} s^{\alpha-1} ds \right) \frac{\pi}{\sin \pi\alpha} \lambda^{2-\alpha} \right] \varphi_h^2 d\lambda \right] dx
\end{aligned} \tag{3.52}
$$

eq: semi energy

As $\mathscr{E}_s \le 0$ under the assumption $a_1 = a_2$, we can take the same supposition as in the continuous case,

$$
\begin{cases}
\displaystyle\int_0^\infty b_1 \frac{\sin \pi\alpha}{\pi} \lambda^{\alpha-1} d\lambda = \frac{a_1}{\epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty})} - a_3, & \text{(3.53a)}\\
\displaystyle a_4 = \left(\frac{\tau^\alpha}{\epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty})} a_1 - b_1 \right) \frac{\pi}{\sin \pi\alpha} \lambda^{1-\alpha}, & \text{(3.53b)}\\
a_1 = a_2, & \text{(3.53c)}\\
\displaystyle a_3 - \int_0^\infty \frac{b_1^2}{a_4} d\lambda \ge 0, & \text{(3.53d)}\\
\displaystyle a_1 M - \int_0^\infty \frac{(a_1 \tau^\alpha M + b_1 \lambda)^2}{4\lambda a_4} d\lambda \ge 0. & \text{(3.53e)}
\end{cases}
$$

As a result, we can choose the parameters a_1, a_2, a_3, a_4, b_1 same as in the continuous case, which statisfies $\frac{d\mathscr{E}}{dt} \le 0$, so we complete the proof of the theorem. □

Remark 3.1. *We can use different kinds of numerical flux, such as [11], we can get the same result as above by similar calculations.*

3.3 The error analysis of semi-discrete DG scheme

The Gauss-Radau projections $P_h^{\pm}$ play an important role in deriving optimal error estimates of DG methods using upwind fluxes. Taking the definition of P_h^- in [5]. For any given function $u \in H^1(I_j)$ and an arbitrary cell $I_j = (x_{j-1/2}, x_{j+1/2})$, the restriction of $P_h^{\pm}u$ to I_j is the unique polynomial in V_h^k satisfying

$$\begin{aligned} \int_{I_j} P_h^{\pm}u(x)v_h dx &= \int_{I_j} u(x)v_h dx, \quad \forall v_h \in V_h^{k-1}, \\ P_h^{\pm}u\left(x_{j+1/2}^{\pm}\right) &= u\left(x_{j+1/2}^{\pm}\right), \end{aligned} \tag{3.54}$$

when $k = 0$, we set $P^{-1}(I_j) = 0$. From the above definition, we can see that the projection $P_h^{\pm}$ can be solved explicitly on each element.

The projection described above satisfies the following approximation error estimates. Let $u \in W^{k+1,p}(I_j)(p = 2, \infty)$, and according to the standard approximation theory [3], we have

$$\|u - P_h^- u\|_{L^p(I_j)} \leq C h_j^{k+1} \|u\|_{W^{k+1,p}(I_j)}, \tag{3.55}$$

where C is independent of h_j.

Remark 3.2. *The variable u is just a scalar, so we can't use P_h^- directly in Cole-Cole model, but we can define a new projection based on P_h^-.*

When using the upwind flux, we introduce the following coupled projection as in [6] $(P_h^{1,\star}E, P_h^{2,\star}H) \in V_h^k \times V_h^k$, which satisfies

$$\begin{aligned} \int_{I_j} P_h^{1,\star}E(x)v_h dx &= \int_{I_j} E(x)v_h dx, \quad \forall v_h \in P^{k-1}(I_j), \\ \int_{I_j} P_h^{2,\star}H(x)v_h dx &= \int_{I_j} H(x)v_h dx, \quad \forall v_h \in P^{k-1}(I_j), \\ \left(\{\{P_h^{1,\star}E\}\} + \frac{1}{2}\sqrt{\frac{\mu_0}{\epsilon_0\epsilon_{r,\infty}}}[\![P_h^{2,\star}H]\!]\right)|_{j-1/2} &= E(x_{j-1/2}), \\ \left(\{\{P_h^{2,\star}H\}\} + \frac{1}{2}\sqrt{\frac{\epsilon_0\epsilon_{r,\infty}}{\mu_0}}[\![P_h^{1,\star}E]\!]\right)|_{j-1/2} &= H(x_{j-1/2}), \end{aligned} \tag{3.56}$$

eq:coupled proj

as a consequence, the following lemma plays an important role in the error analysis of the DG scheme [6]

Lemma 3.1. *The projection* (3.56) *is well-defined and satisfies*

$$\begin{aligned} P_h^{1,\star}E &= \frac{1}{2}\left\{P_h^+\left(E+\sqrt{\frac{\mu_0}{\epsilon_0\epsilon_{r,\infty}}}H\right)+P_h^-\left(E-\sqrt{\frac{\mu_0}{\epsilon_0\epsilon_{r,\infty}}}H\right)\right\},\\ P_h^{2,\star}H &= \frac{1}{2}\left\{P_h^+\left(H+\sqrt{\frac{\epsilon_0\epsilon_{r,\infty}}{\mu_0}}E\right)+P_h^-\left(H-\sqrt{\frac{\epsilon_0\epsilon_{r,\infty}}{\mu_0}}E\right)\right\}, \end{aligned} \tag{3.57}$$

and in particular, it satisfies

$$\|P_h^{1,\star}E-E\|_{I_j}\le Ch^{k+1},\quad \|P_h^{2,\star}H-H\|_{I_j}\le Ch^{k+1}, \tag{3.58}$$

new projection

where C is independent of h.

Remark 3.3. *The proof of the lemma is similar to the proof in [6], and it figures out the reason why we choose this kind of projection in [14].*

Remark 3.4. *In the proof of optimal error, we use $P_h^{1,\star}$ and $P_h^{2,\star}$ for electric field E and magnetic field H respectively, and we use L^2 projection P_h^2 for the polarization P and the auxiliary variable φ for fractional derivative.*

Theorem 3.2. *Let $(E,H,P,\varphi)\in H^1(0,T;L^2(\Omega))$ be the exact solutions of* (3.2)*, and (E_h,H_h,P_h,φ_h) be the numerical solutions of semi-discrete DG scheme* (3.38a)-(3.38d)*. The following optimal error estimate holds true:*

$$\|E(t)-E_h(t)\|^2_{L^2(\Omega)}+\|H(t)-H_h(t)\|^2_{L^2(\Omega)}+\|P(t)-P_h(t)\|^2_{L^2(\Omega)}\le Ch^{2k+2},\quad \forall t\ge 0, \tag{3.59}$$

where C depends on $\Omega,k,t,\epsilon_{r,s},\epsilon_{r,\infty},\sigma_{\max}$.

证明. Using the definition of projections, we can decompose the errors as follows:

$$\begin{aligned} E-E_h&=\xi_E-\delta_E,\quad \xi_E\triangleq P_h^{1,\star}E-E_h,\quad \delta_E\triangleq P_h^{1,\star}E-E,\\ H-H_h&=\xi_H-\delta_H,\quad \xi_H\triangleq P_h^{2,\star}H-H_h,\quad \delta_H\triangleq P_h^{2,\star}H-H,\\ P-P_h&=\xi_P-\delta_P,\quad \xi_P\triangleq P_h^{2}P-P_h,\quad \delta_P\triangleq P_h^{2}P-P,\\ \varphi-\varphi_h&=\xi_\varphi-\delta_\varphi,\quad \xi_\varphi\triangleq P_h^{2}\varphi-\varphi_h,\quad \delta_\varphi\triangleq P_h^{2}\varphi-\varphi. \end{aligned} \tag{3.60}$$

eq:error decomp

the exact solutions (E,H,P,φ) also satisfy the semi-discrete scheme (3.38a)-(3.38d) with the same numerical fluxes. As a consequence, we use the semi-scheme for numerical solu-

tions and exact solutions, besides (3.60), we have

$$\begin{aligned}&\int_{I_j}\left[\mu_0\partial_t\xi_H v_h+\xi_E\partial_x v_h+\frac{\mu_0}{\epsilon_0}\sigma_x\xi_H v_h\right]dx-\widehat{\xi_E}v_h^-|_{j+1/2}+\widehat{\xi_E}v_h^+|_{j-1/2}\\=&\int_{I_j}\left[\mu_0\partial_t\delta_H v_h+\delta_E\partial_x v_h+\frac{\mu_0}{\epsilon_0}\sigma_x\delta_H v_h\right]dx-\widehat{\delta_E}v_h^-|_{j+1/2}+\widehat{\delta_E}v_h^+|_{j-1/2}\end{aligned}\tag{3.61}$$

$$\begin{aligned}&\int_{I_j}\left[\epsilon_0\epsilon_{r,\infty}\partial_t\xi_E v_h+\xi_H\partial_x v_h+\partial_t\xi_P v_h+\sigma_x\epsilon_{r,\infty}\xi_E v_h+\frac{\sigma_x}{\epsilon_0}\xi_P v_h\right]dx-\widehat{\xi_H}v_h^-|_{j+1/2}+\widehat{\xi_H}v_h^+|_{j-1/2}\\=&\int_{I_j}\left[\epsilon_0\epsilon_{r,\infty}\partial_t\delta_E v_h+\delta_H\partial_x v_h+\partial_t\delta_P v_h+\sigma_x\epsilon_{r,\infty}\delta_E v_h+\frac{\sigma_x}{\epsilon_0}\delta_P v_h\right]dx-\widehat{\delta_H}v_h^-|_{j+1/2}+\widehat{\delta_H}v_h^+|_{j-1/2}\end{aligned}\tag{3.62}$$

$$\begin{aligned}&\int_{I_j}\left[\xi_P v_h+\int_0^\infty\tau^\alpha\xi_\varphi v_h d\lambda-\epsilon_0(\epsilon_{r,s}-\epsilon_{r,\infty})\xi_E v_h\right]dx\\=&\int_{I_j}\left[\delta_P v_h+\int_0^\infty\tau^\alpha\delta_\varphi v_h d\lambda-\epsilon_0(\epsilon_{r,s}-\epsilon_{r,\infty})\delta_E v_h\right]dx\end{aligned}\tag{3.63}$$

$$\begin{aligned}&\int_{I_j}\left[\partial_t\xi_\varphi v_h+\lambda\xi_\varphi v_h-\frac{\sin\pi\alpha}{\pi}\lambda^{\alpha-1}\partial_t\xi_P v_h\right]dx\\=&\int_{I_j}\left[\partial_t\delta_\varphi v_h+\lambda\delta_\varphi v_h-\frac{\sin\pi\alpha}{\pi}\lambda^{\alpha-1}\partial_t\delta_P v_h\right]dx\end{aligned}\tag{3.64}$$

For energy we defined in last section, we can derive the similar steps as following

$$\frac{d}{dt}\mathscr{E}(\xi_E,\xi_H,\xi_P,\xi_\varphi)=\mathscr{E}_{\xi,1}+\mathscr{E}_{\xi,2},\tag{3.65}$$

eq: energy for

where

$$
\begin{aligned}
\mathscr{E}_{\xi,1} = &- \int_\Omega a_1 \sigma_x \epsilon_{r,\infty} \xi_E^2 dx - \int_\Omega a_2 \frac{\mu_0}{\epsilon_0} \sigma_x \xi_H^2 dx - \int_\Omega a_1 \frac{\sigma_x}{\epsilon_0} \frac{1}{\epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty})} \xi_P^2 dx \\
&+ \int_\Omega \left(- \frac{a_1}{\epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty})} + a_3 + \int_0^\infty b_1 \frac{\sin \pi\alpha}{\pi} \lambda^{\alpha-1} d\lambda \right) \epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty}) \xi_E \partial_t \xi_P dx \\
&- \int_\Omega \int_0^\infty \left(a_1 \frac{\sigma_x}{\epsilon_0} \frac{\tau^\alpha}{\epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty})} + b_1 \lambda \right) \xi_\varphi \xi_P d\lambda dx \\
&- \int_\Omega \int_0^\infty \left[-b_1 \frac{\pi}{\sin \pi\alpha} \lambda^{2-\alpha} + \tau^\alpha \left(a_3 + \int_0^\infty b_1 \frac{\sin \pi\alpha}{\pi} s^{\alpha-1} ds \right) \frac{\pi}{\sin \pi\alpha} \lambda^{2-\alpha} \right] \xi_\varphi^2 d\lambda dx \\
&+ \int_\Omega \int_0^\infty \left[a_4 + b_1 \frac{\pi}{\sin \pi\alpha} \lambda^{1-\alpha} - \tau^\alpha \left(a_3 + \int_0^\infty b_1 \frac{\sin \pi\alpha}{\pi} s^{\alpha-1} ds \right) \frac{\pi}{\sin \pi\alpha} \lambda^{1-\alpha} \right] \xi_\varphi \partial_t \xi_\varphi d\lambda dx \\
&- \sum_{j=1}^M \int_\Omega \left(a_1 \xi_H \partial_x \xi_E + a_2 \xi_E \partial_x \xi_H \right) dx \\
&+ \sum_{j=1}^M a_1 \widehat{\xi_H} \xi_E^- |_{j+1/2} - a_1 \widehat{\xi_H} \xi_E^+ |_{j-1/2} + a_2 \widehat{\xi_E} \xi_H^- |_{j+1/2} - a_2 \widehat{\xi_E} \xi_H^+ |_{j-1/2},
\end{aligned}
\tag{3.66}
$$

eq: E_{xi,1}

and

$$
\begin{aligned}
\mathscr{E}_{\xi,2} =& \sum_{j=1}^{M} \int_{I_j} \left(a_1 \epsilon_0 \epsilon_{r,\infty} \partial_t \delta_E \xi_E + a_1 \partial_t \delta_P \xi_E + a_1 \sigma_x \epsilon_{r,\infty} \delta_E \xi_E + a_1 \frac{\sigma_x}{\epsilon_0} \delta_P \xi_E \right) dx \\
&+ \int_{I_j} a_1 \delta_H \partial_x \xi_E dx - a_1 \widehat{\delta_H} \xi_E^-|_{j+1/2} + a_1 \widehat{\delta_H} \xi_E^+|_{j-1/2} \\
&+ \sum_{j=1}^{M} \int_{I_j} \left(a_2 \mu_0 \partial_t \delta_H \xi_H + a_2 \frac{\mu_0}{\epsilon_0} \sigma_x \delta_H \xi_H \right) dx + \int_{I_j} a_2 \delta_E \partial_x \xi_H dx - a_2 \widehat{\delta_E} \xi_H^-|_{j+1/2} + a_2 \widehat{\delta_E} \xi_H^+|_{j-1/2} \\
&+ \sum_{j=1}^{M} \int_{I_j} \int_0^{\infty} b_1 \left(\partial_t \delta_\varphi \xi_P + \lambda \delta_\varphi \xi_P - \frac{\sin \pi\alpha}{\pi} \lambda^{\alpha-1} \partial_t \delta_P \xi_P \right) d\lambda dx \\
&+ \sum_{j=1}^{M} \int_{I_j} \int_0^{\infty} b_1 \frac{\pi}{\sin \pi\alpha} \lambda^{1-\alpha} \left(-\partial_t \delta_\varphi \xi_\varphi - \lambda \delta_\varphi \xi_\varphi + \frac{\sin \pi\alpha}{\pi} \lambda^{\alpha-1} \partial_t \delta_P \xi_\varphi \right) d\lambda dx \\
&+ \sum_{j=1}^{M} \int_{I_j} a_1 \frac{\sigma_x}{\epsilon_0} \frac{1}{\epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty})} \left(\delta_P \xi_P + \tau^\alpha \int_0^{\infty} \delta_\varphi \xi_P d\lambda - \epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty}) \delta_E \xi_P \right) dx \\
&+ \sum_{j=1}^{M} \int_{I_j} \frac{a_1}{\epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty})} \left(\delta_P \partial_t \xi_P + \tau^\alpha \int_0^{\infty} \delta_\varphi \partial_t \xi_P d\lambda - \epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty}) \delta_E \partial_t \xi_P \right) dx \\
&+ \sum_{j=1}^{M} \int_{I_j} \frac{a_1}{\epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty})} \left(- \int_0^{\infty} \tau^\alpha \frac{\pi}{\sin \pi\alpha} \lambda^{1-\alpha} \left(-\partial_t \delta_\varphi \xi_\varphi - \lambda \delta_\varphi \xi_\varphi + \frac{\sin \pi\alpha}{\pi} \lambda^{\alpha-1} \partial_t \delta_P \xi_\varphi \right) d\lambda \right) dx
\end{aligned}
\tag{3.67}
$$

eq: E_{xi,2}

As the projections we defined in (3.56) are linear operator, thus $\widehat{\xi_E}, \widehat{\xi_H}$ have the same form of $\widehat{E_h}, \widehat{H_h}$ respectively. As a consequence, we can handle (3.66) in the same way of Theorem 3.1 and then get the following

$$
\mathscr{E}_{\xi,1} = \mathscr{E}_s \left(\xi_E, \xi_H, \xi_P, \xi_\varphi \right) + \mathscr{E}_{sn}, \tag{3.68}
$$

where $\mathscr{E}_s \left(\xi_E, \xi_H, \xi_P, \xi_\varphi \right)$ has the same form of $\mathscr{E}_s$ we defined above, and $\mathscr{E}_{sn}$ is the terms in (3.52) except $\mathscr{E}_s$ and the variables are replaced by $\xi_E, \xi_H, \xi_P, \xi_\varphi$, which is keeping semi-negative definite with choosing the same a_1, a_2, a_3, a_4, b_1 as above. So it's obviously that $\mathscr{E}_{\xi,1} \leq 0$.

For terms in $\mathscr{E}_{\xi,2}$, the PML parameter is bounded, i.e. $0 \leq \sigma_x \leq \sigma_{\max}$ by definition. We depart $\mathscr{E}_{\xi,2}$ as following

$$
\mathscr{E}_{\xi,2} = \sum_{i=1}^{4} G_i, \tag{3.69}
$$

where

$$
\begin{aligned}
G_1 &= \int_\Omega \left(a_1\epsilon_0\epsilon_{r,\infty}\partial_t\delta_E\xi_E + a_1\partial_t\delta_P\xi_E + a_1\sigma_x\epsilon_{r,\infty}\delta_E\xi_E + a_1\frac{\sigma_x}{\epsilon_0}\delta_P\xi_E \right) dx \\
&\quad + \int_\Omega \left(a_2\mu_0\partial_t\delta_H\xi_H + a_2\frac{\mu_0}{\epsilon_0}\sigma_x\delta_H\xi_H \right) dx \\
G_2 &= \sum_{j=1}^{M}\int_{I_j} a_1\delta_H\partial_x\xi_E dx - a_1\widehat{\delta_H}\xi_E^-|_{j+1/2} + a_1\widehat{\delta_H}\xi_E^+|_{j-1/2} \\
&\quad + \sum_{j=1}^{M}\int_{I_j} a_2\delta_E\partial_x\xi_H dx - a_2\widehat{\delta_E}\xi_H^-|_{j+1/2} + a_2\widehat{\delta_E}\xi_H^+|_{j-1/2} \\
G_3 &= -\int_\Omega a_1\frac{\sigma_x}{\epsilon_0}\delta_E\xi_P + a_1\delta_E\partial_t\xi_P dx \\
G_4 &= \mathscr{E}_{\xi,2} - G_1 - G_2 - G_3.
\end{aligned}
\tag{3.70}
$$

Using the Young's inequality, and the estimate in Lemma 3.1

$$
\begin{aligned}
G_1 &\le \int_\Omega \frac{a_1}{2}\epsilon_0\epsilon_{r,\infty}\xi_E^2 + \frac{a_2}{2}\mu_0\xi_H^2 dx + \int_\Omega a_2\mu_0(\partial_t\delta_H)^2 + a_2\mu_0\left(\frac{\sigma_x}{\epsilon_0}\right)^2\delta_H^2 dx \\
&\quad + \int_\Omega 2a_1\epsilon_0\epsilon_{r,\infty}(\partial_t\delta_E)^2 + \frac{2a_1}{\epsilon_0\epsilon_{r,\infty}}(\partial_t\delta_P)^2 + \frac{2a_1\epsilon_{r,\infty}}{\epsilon_0}\sigma_x^2\delta_E^2 + \frac{2a_1}{\epsilon_0^3\epsilon_{r,\infty}}\delta_P^2 dx \\
&\le \int_\Omega \left(\frac{a_1}{2}\epsilon_0\epsilon_{r,\infty}\xi_E^2 + \frac{a_2}{2}\mu_0\xi_H^2\right) dx + C_1 h^{2k+2}
\end{aligned}
\tag{3.71}
$$

where C_1 depends on $\Omega, k, t, \epsilon_{r,\infty}, \sigma_{\max}, a_1, a_2$.

For G_2, by the continuity of E, H, we have $\widehat{\delta_E} = \widehat{\delta_H} = 0$. And $\xi_E, \xi_H \in V_h$, so $\partial_x\xi_E, \partial_x\xi_H \in \mathcal{P}^{k-1}$, and by the definition of projecitons (3.56), $G_2 = 0$.

Integrating G_3 in $[0,t]$, and $\xi_P(x,0) = 0$ from the initial condition, we have

$$
\begin{aligned}
\int_0^t G_3 d\tau &= \int_0^t\int_\Omega a_1\partial_t\delta_E\xi_P dx d\tau - \int_0^t\int_\Omega a_1\frac{\sigma_x}{\epsilon_0}\delta_E\xi_P dx d\tau - \int_\Omega a_1\delta_E\xi_P dx \\
&\le \int_0^t\int_\Omega k_1\xi_P^2 + \frac{a_1^2}{2k_1}(\partial_t\delta_E)^2 + \frac{a_1^2\sigma_x^2}{2\epsilon_0^2 k_1}\delta_E^2 dx d\tau + \int_\Omega k_1\xi_P^2 + \frac{a_1^2}{2k_1}\delta_E^2 dx \\
&\le \int_0^t\int_\Omega k_1\xi_P^2 dx d\tau + \int_\Omega k_1\xi_P^2 dx + C_2 h^{2k+2}
\end{aligned}
\tag{3.72}
$$

where $k_1 = \frac{a_1}{4\epsilon_0(\epsilon_{r,s}-\epsilon_{r,\infty})}$, and C_2 depends on $\Omega, k, t, \epsilon_{r,s}, \epsilon_{r,\infty}, \sigma_{\max}, a_1$.

We notice that the terms in G_4 only contain the cross terms of $\delta_P, \delta_\phi, \xi_P, \xi_\varphi$ and their time derivatives, thus by the definition of L^2 projection P_h^2 and the domain of ξ_P, ξ_φ, we have $G_4 = 0$ immediately.

Now integrating (3.65) in $[0,t]$, and combing the above estimates, we have

$$\begin{aligned}\mathscr{E}(\xi_E(t),\xi_H(t),\xi_P(t),\xi_\varphi(t)) \le \mathscr{E}(\xi_E(0),\xi_H(0),\xi_P(0),\xi_\varphi(0)) + C_3h^{2k+2} \\ + \int_0^t\int_\Omega \left(\frac{a_1}{2}\epsilon_0\epsilon_{r,\infty}\xi_E^2 + \frac{a_2}{2}\mu_0\xi_H^2 + k_1\xi_P^2\right)dxd\tau + \int_\Omega k_1\xi_P^2dx,\end{aligned} \tag{3.73}$$

eq: energy esti

where C_3 depends on $\Omega, k, t, \epsilon_{r,s}, \epsilon_{r,\infty}, \sigma_{\max}, a_1, a_2$, and by the form of $\mathscr{E}$ and k_1, we have

$$\begin{aligned}\tilde{\mathscr{E}}(\xi_E(t),\xi_H(t),\xi_P(t)) \le \mathscr{E}(\xi_E(0),\xi_H(0),\xi_P(0),\xi_\varphi(0)) + C_3h^{2k+2} \\ + \int_0^t \tilde{\mathscr{E}}(\xi_E(\tau),\xi_H(\tau),\xi_P(\tau))d\tau,\end{aligned} \tag{3.74}$$

where

$$\tilde{\mathscr{E}}(\xi_E(t),\xi_H(t),\xi_P(t)) = \int_\Omega \left(\frac{a_1}{2}\epsilon_0\epsilon_{r,\infty}\xi_E^2 + \frac{a_2}{2}\mu_0\xi_H^2 + \frac{a_1}{4\epsilon_0(\epsilon_{r,s}-\epsilon_{r,\infty})}\xi_P^2\right)dx, \tag{3.75}$$

and for simplicity, we choose $O(h^{k+1})$ initial approximation:

$$E_h(0) = P_h^{1,\star}E(0), H_h(0) = P_h^{2,\star}H(0), P_h(0) = P_h^2P(0), \varphi_h(0) = P_h^2\varphi(0), \tag{3.76}$$

so we have

$$\tilde{\mathscr{E}}(\xi_E(t),\xi_H(t),\xi_P(t)) \le Ch^{2k+2} + \int_0^t \tilde{\mathscr{E}}(\xi_E(\tau),\xi_H(\tau),\xi_P(\tau))d\tau, \tag{3.77}$$

and by the Grönwall's inequality

$$\tilde{\mathscr{E}}(\xi_E(t),\xi_H(t),\xi_P(t)) \le Ch^{2k+2}, \tag{3.78}$$

where C depends on $\Omega, k, t, \epsilon_{r,s}, \epsilon_{r,\infty}, \sigma_{\max}, a_1, a_2$. And by the triangle inequality and the estimates of $\delta_E, \delta_H, \delta_P$, we complete the proof. □

Remark 3.5. *In* (3.73)*, we move the last term in the right to the left, we can write as following*

$$\int_\Omega \frac{a_1}{2}\epsilon_0\epsilon_{r,\infty}\xi_E^2 + \frac{a_2}{2}\mu_0\xi_H^2 + \left(\frac{a_3}{2} - k_1 - \frac{1}{2}\int_0^\infty \frac{b_1^2}{a_4}d\lambda\right)\xi_P^2 + \frac{1}{2}\int_0^\infty \left(\frac{b_1}{\sqrt{a_4}}\xi_P + \sqrt{a_4}\xi_\varphi\right)^2 d\lambda dx, \tag{3.79}$$

which is greater than

$$\int_\Omega \frac{a_1}{2}\epsilon_0\epsilon_{r,\infty}\xi_E^2 + \frac{a_2}{2}\mu_0\xi_H^2 + \left(\frac{a_3}{2} - k_1 - \frac{1}{2}\int_0^\infty \frac{b_1^2}{a_4}d\lambda\right)\xi_P^2dx. \tag{3.80}$$

From the choice of a_1, a_2, a_3, a_4, b_1*, we can see that*

$$\frac{a_3}{2} - \frac{1}{2}\int_0^\infty \frac{b_1^2}{a_4} d\lambda = \frac{a_1}{2\epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty})}, \tag{3.81}$$

to ensure the consistency of the integrand in the spatial integrals over Ω *at both ends, we choose* $k_1 = \frac{a_1}{4\epsilon_0(\epsilon_{r,s}-\epsilon_{r,\infty})}$.

4 Fully discrete scheme in 1D

Since the Cole-Cole model is characterized by the fractional derivative, its solution generally exhibits weak regularity at initial time $t = 0$, which always deteriorates the convergence accuracy [12, 15].

4.1 Fully discrete scheme for energy dissipation

We already have the semi-discrete energy dissipation in the previous section. To construct the fully discrete scheme, we divide the time interval $[0, T]$ into uniform subintervals using points $0 = t_0 < t_1 < \cdots < t_M = T$, where $t_m = m\tau_0$. Futhermore, we denote $u^m = u(\cdot, m\tau_0)$ and the following finite difference operators:

$$\delta_{\tau_0} u^m = \frac{u^m - u^{m-1}}{\tau_0}, \quad \overline{u}^m = \frac{u^m + u^{m-1}}{2}.$$

We can formulate a Crank-Nickolson type difference scheme for the semi-discrete scheme(3.38a)-(3.38d): given the initial approximations $E_h^0, H_h^0, P_h^0 = 0, \varphi_h^0 = 0$, for all

$k \geq 1$ find $E_h^k, H_h^k, P_h^k, \varphi_h^k \in V_h$, and test function $v \in V_h$ such that

$$\int_{I_j} \mu_0 \delta_{\tau_0} H_h^k v = \widehat{\overline{E}_h^k} v^-|_{j+1/2} - \widehat{\overline{E}_h^k} v^+|_{j-1/2} - \int_{I_j} \overline{E}_h^k \frac{\partial v}{\partial x} dx - \int_{I_j} \frac{\mu_0}{\epsilon_0} \sigma_x \overline{H}_h^k v dx, \tag{4.1a}$$

fully H

$$\begin{aligned} \int_{I_j} \epsilon_0 \epsilon_{r,\infty} \delta_{\tau_0} E_h^k v dx =& \widehat{\overline{H}_h^k} v^-|_{j+1/2} - \widehat{\overline{H}_h^k} v^+|_{j-1/2} - \int_{I_j} \overline{H}_h^k \frac{\partial v}{\partial x} dx \\ & - \int_{I_j} \delta_{\tau_0} P_h^k v dx - \int_{I_j} \sigma_x \epsilon_{r,\infty} \overline{E}_h^k v dx - \int_{I_j} \frac{\sigma_x}{\epsilon_0} \overline{P}_h^k v dx, \end{aligned} \tag{4.1b}$$

fully E

$$\int_{I_j} \overline{P}_h^k v dx + \int_{I_j} \int_0^\infty \tau^\alpha \overline{\varphi}_h^k v d\lambda dx = \int_{I_j} \epsilon_0 (\epsilon_{r,s} - \epsilon_{r,\infty}) \overline{E}_h^k v dx, \tag{4.1c}$$

fully P

$$\int_{I_j} \delta_{\tau_0} \varphi_h^k v dx = -\lambda \int_{I_j} \overline{\varphi}_h^k v dx + \frac{\sin(\pi\alpha)}{\pi} \lambda^{\alpha-1} \int_{I_j} \delta_{\tau_0} P_h^k v dx, \tag{4.1d}$$

fully varphi

where the numerical fluxes keep the same form as in the semi-discrete scheme. In practical implementation, the scheme (4.1a)-(4.1d) can be realized as follows: First, from (4.1c), we represent P_h^k using E_h^k and φ_h^k; then substitute P_h^k into (4.1d), and using E_h^k to solve φ_h^k, and taking back to (4.1c) to get P_h^k; finally, substitute P_h^k into (4.1b) and solving the resulting equation along with (4.1a) for both E_h^k and H_h^k. The solvability of the system for both E_h^k and H_h^k can be proved in the same way as in our previous work [8]. Why need to explain the solvability?

Theorem 4.1. *For the solutions $E_h^k, H_h^k, P_h^k, \varphi_h^k (k \geq 1)$ of the scheme (4.1a)-(4.1d), we have the fully discrete stability:*

$$\mathscr{E}(E_h^k, H_h^k, P_h^k, \varphi_h^k) \leq \mathscr{E}(E_h^{k-1}, H_h^{k-1}, P_h^{k-1}, \varphi_h^{k-1}), \tag{4.2}$$

where $\mathscr{E}$ has the same definition as in (3.49), for simplicity we denote $\mathscr{E}_h^k := \mathscr{E}(E_h^k, H_h^k, P_h^k, \varphi_h^k)$.

证明. Choosing $v = \frac{\tau_0}{2}\left(H_h^k + H_h^{k-1}\right)$ in (4.1a), $v = \frac{\tau_0}{2}\left(E_h^k + E_h^{k-1}\right)$ in (4.1b), then the

resulting equation for H_h^k becomes to

$$\int_{I_j} \mu_0 \frac{H_h^k - H_h^{k-1}}{\tau_0} \frac{\tau_0 \left(H_h^k + H_h^{k-1}\right)}{2} dx = \tau_0 \widehat{\overline{E}_h^k} \overline{H}_h^{k,-}|_{j+1/2} - \tau_0 \widehat{\overline{E}_h^k} \overline{H}_h^{k,+}|_{j-1/2} - \int_{I_j} \tau_0 \overline{E}_h^k \frac{\partial \overline{H}_h^k}{\partial x} dx - \int_{I_j} \frac{\mu_0}{\epsilon_0} \sigma_x \tau_0 \left(\overline{H}_h^k\right)^2 dx, \tag{4.3}$$

and the resulting equation for E_h^k becomes to

$$\begin{aligned} \int_{I_j} \epsilon_0 \epsilon_{r,\infty} \frac{E_h^k - E_h^{k-1}}{\tau_0} \frac{\tau_0 \left(E_h^k + E_h^{k-1}\right)}{2} dx =& \tau_0 \widehat{\overline{H}_h^k} \overline{E}_h^{k,-}|_{j+1/2} - \tau_0 \widehat{\overline{H}_h^k} \overline{E}_h^{k,+}|_{j-1/2} - \int_{I_j} \tau_0 \overline{H}_h^k \frac{\partial \overline{E}_h^k}{\partial x} dx \\ & - \int_{I_j} \frac{P_h^k - P_h^{k-1}}{\tau_0} \frac{\tau_0 \left(E_h^k + E_h^{k-1}\right)}{2} dx \\ & - \int_{I_j} \sigma_x \epsilon_{r,\infty} \tau_0 \left(\overline{E}_h^k\right)^2 dx - \int_{I_j} \frac{\sigma_x}{\epsilon_0} \tau_0 \overline{P}_h^k \overline{E}_h^k dx. \end{aligned} \tag{4.4}$$

Firstly, we consider the spatial part of the fully discrete scheme, which is similar to the semi-discrete scheme. We can get the similar equation, because the spatial part is independent of the time discretization. As a result the spatial part of the fully discrete energy function is reduced to

$$\begin{aligned} \mathscr{E}_s^k = & - \sum_{j=1}^{M-1} \mathscr{M}_{j+1/2}(\overline{E}_h^k, \overline{H}_h^k) - \widehat{\overline{E}_{1/2}^k} \overline{H}_{1/2}^{k,+} - \widehat{\overline{H}_{1/2}^k} \overline{E}_{1/2}^{k,+} + \overline{E}_{1/2}^{k,+} \overline{H}_{1/2}^{k,+} \\ & + \widehat{\overline{E}_{M+1/2}^k} \overline{H}_{M+1/2}^{k,-} + \widehat{\overline{H}_{M+1/2}^k} \overline{E}_{M+1/2}^{k,-} - \overline{E}_{M+1/2}^{k,-} \overline{H}_{M+1/2}^{k,-}, \end{aligned} \tag{4.5}$$

with the same choice of the numerical fluxes as in the semi-discrete scheme.

$$\mathscr{E}_s^k = - \sum_{j=1}^{M-1} \mathscr{M}_{j+1/2}(\overline{E}_h^k, \overline{H}_h^k) - \sqrt{\frac{\epsilon_0 \epsilon_{r,\infty}}{\mu_0}} \left(\overline{E}_{1/2}^{k,+}\right)^2 - \sqrt{\frac{\epsilon_0 \epsilon_{r,\infty}}{\mu_0}} \left(\overline{H}_{M+1/2}^{k,-}\right)^2 \leq 0. \tag{4.6}$$

Now, we do time difference instead of the time derivative for the energy function. We recall the definition of the energy function in fully-discrete scheme

$$\mathscr{E}_h^k = \int_\Omega \left(\frac{a_1}{2} \epsilon_0 \epsilon_{r,\infty} \left(E_h^k\right)^2 + \frac{a_2}{2} \mu_0 \left(H_h^k\right)^2 + \frac{a_3}{2} \left(P_h^k\right)^2 + \frac{1}{2} \int_0^\infty a_4 \left(\varphi_h^k\right)^2 d\lambda + \int_0^\infty b_1 \varphi_h^k P_h^k d\lambda \right) dx, \tag{4.7}$$

eq: definition

in order to keep the positiveness of the energy, it's the same with the semi-discrete scheme, where $\mathcal{E}_h^k \geq 0$ is equivalent to

$$a_3 - \int_0^\infty \frac{b_1^2}{a_4} d\lambda \geq 0. \tag{4.8}$$

The time difference of the energy function in the fully discrete scheme is

$$\begin{aligned}
&\int_\Omega \frac{a_1}{2}\epsilon_0\epsilon_{r,\infty}\frac{\left(E_h^k\right)^2 - \left(E_h^{k-1}\right)^2}{\tau_0} + \frac{a_2}{2}\mu_0\frac{\left(H_h^k\right)^2 - \left(H_h^{k-1}\right)^2}{\tau_0} + \frac{a_3}{2}\frac{\left(P_h^k\right)^2 - \left(P_h^{k-1}\right)^2}{\tau_0} \\
&+ \int_0^\infty \frac{a_4}{2}\frac{\left(\varphi_h^k\right)^2 - \left(\varphi_h^{k-1}\right)^2}{\tau_0} d\lambda + \int_0^\infty b_1 \frac{P_h^k\varphi_h^k - P_h^{k-1}\varphi_h^{k-1}}{\tau_0} d\lambda \quad dx \\
=&\int_\Omega \mathscr{E}_s^k - a_1\delta_{\tau_0}P_h^k\overline{E}_h^k - a_1\sigma_x\epsilon_{r,\infty}\left(\overline{E}_h^k\right)^2 - a_1\frac{\sigma_x}{\epsilon_0}\overline{P}_h^k\overline{E}_h^k - a_2\frac{\mu_0}{\epsilon_0}\sigma_x\left(\overline{H}_h^k\right)^2 - a_3\tau^\alpha\int_0^\infty \overline{\varphi}_h^k\delta_{\tau_0}P_h^k d\lambda \\
&+ a_3\epsilon_0\left(\epsilon_{r,s} - \epsilon_{r,\infty}\right)\overline{E}_h^k\delta_{\tau_0}P_h^k - \int_0^\infty a_4\lambda\left(\overline{\varphi}_h^k\right)^2 + a_4\frac{\sin\pi\alpha}{\pi}\lambda^{\alpha-1}\delta_{\tau_0}P_h^k\overline{\varphi}_h^k d\lambda \\
&+ \int_0^\infty b_1\left(\overline{P}_h^k\delta_{\tau_0}\varphi_h^k + \delta_{\tau_0}P_h^k\overline{\varphi}_h^k\right) d\lambda \quad dx \\
=&\int_\Omega \mathscr{E}_s^k - a_1\sigma_x\epsilon_{r,\infty}\left(\overline{E}_h^k\right)^2 - a_2\frac{\mu_0}{\epsilon_0}\sigma_x\left(\overline{H}_h^k\right)^2 - a_1\frac{\sigma_x}{\epsilon_0}\frac{1}{\epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty})}\left(\overline{P}_h^k\right)^2 \\
&- \epsilon_0\left(\epsilon_{r,s} - \epsilon_{r,\infty}\right)\left[\frac{a_1}{\epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty})} - a_3 - \int_0^\infty b_1\frac{\sin\pi\alpha}{\pi}\lambda^{\alpha-1}d\lambda\right]\delta_{\tau_0}P_h^k\overline{E}_h^k \\
&- \int_0^\infty \left(a_1\frac{\sigma_x}{\epsilon_0}\frac{\tau_0}{\epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty})} + b_1\lambda\right)\overline{P}_h^k\overline{\varphi}_h^k d\lambda \\
&+ \int_0^\infty \left[a_4 + b_1\frac{\pi}{\sin\pi\alpha}\lambda^{1-\alpha} - \tau^\alpha\left(a_3 + \int_0^\infty b_1\frac{\sin\pi\alpha}{\pi}\lambda^{\alpha-1}d\lambda\right)\frac{\pi}{\sin\pi\alpha}\lambda^{1-\alpha}\right]\delta_{\tau_0}\varphi_h^k\overline{\varphi}_h^k d\lambda \\
&- \int_0^\infty \left[-\frac{\pi}{\sin\pi\alpha}b_1\lambda^{2-\alpha} + \tau^\alpha\left(a_3 + \int_0^\infty b_1\frac{\sin\pi\alpha}{\pi}\lambda^{\alpha-1}d\lambda\right)\frac{\pi}{\sin\pi\alpha}\lambda^{2-\alpha}\right]\left(\overline{\varphi}_h^k\right)^2 d\lambda \quad dx.
\end{aligned} \tag{4.9}$$

As $\mathscr{E}_s^k \leq 0$ under the assumption $a_1 = a_2$, we can take the same supposition as in the

semi-discrete scheme,

$$\begin{cases} \displaystyle\int_0^\infty b_1 \frac{\sin \pi\alpha}{\pi} \lambda^{\alpha-1} d\lambda = \frac{a_1}{\epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty})} - a_3, & \text{(4.10a)} \\ \displaystyle a_4 = (\frac{\tau^\alpha}{\epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty})} a_1 - b_1) \frac{\pi}{\sin \pi\alpha} \lambda^{1-\alpha}, & \text{(4.10b)} \\ a_1 = a_2, & \text{(4.10c)} \\ \displaystyle a_3 - \int_0^\infty \frac{b_1^2}{a_4} d\lambda \geq 0, & \text{(4.10d)} \\ \displaystyle a_1 M - \int_0^\infty \frac{(a_1 \tau^\alpha M + b_1 \lambda)^2}{4\lambda a_4} d\lambda \geq 0. & \text{(4.10e)} \end{cases}$$

As a result, we can choose the parameters a_1, a_2, a_3, a_4, b_1 same as in the semi-discrete scheme, which is also the same as in continuous case. Hence, the time difference of the energy function is decreasing, which completes the proof.

□

Remark 4.1. *Not as in [10], where fully discrete scheme is stable with a summation term* $\sum_{k=1}^n ||P_h^k - P_h^{k-1}||_0^2$. *The term contains the time difference of the induced polarization in the whole time interval, which is obviously nonlocal, so it's difficult to show the monotonically decreasing of the fully discrete energy function.*

However, the stability analysis presented here does not rely on such a summation term. Instead, we directly analyze a new fully discrete energy function $\mathscr{E}_h^k$ *and its time difference, ensuring that the energy dissipation property holds at each time step. This approach simplifies the proof and avoids the need for additional assumptions on the time difference of the induced polarization.*

And to our knowledge, this is the first time that the fully discrete scheme for the Cole-Cole model is proved to be stable with the monotonical dissipation of the energy function. And the scheme is unconditionally stable.

4.2 The error analysis of fully discrete scheme

Now we consider the same Crank-Nickolson type

5 Stability Analysis in 2D

需要分析二维 Cole-Cole 介质中带有 UPML 吸收边界层的 Maxwell 方程组的稳定性，首先需要对这个方程组引入能量函数。

先将上一小节中方程写做两个方程组，一个关于变量 $\epsilon_{r,\infty}E_z, \mu_r H_x, \mu_r H_y$ 的守恒形，一个关于变量 $P, P_{z,1}, P_{z,2}, P_{z,3}, Q_x, Q_y$ 的广义状态方程 (带有分数阶导数，故称其为广义)。

对于守恒形部分，满足方程

$$\frac{\partial \mathbf{U}^{(1)}}{\partial t} + \nabla \cdot (\mathbb{A}\mathbf{U}^{(1)}) = \mathbf{S}^{(1)}, \tag{5.1}$$

conserve part f

其中 $\mathbf{U}^{(1)} = (\epsilon_{r,s}E_z, \mu_r H_x, \mu_r H_y)^T$， $\mathbf{U}^{(2)} = (P, P_{z,1}, P_{z,2}, P_{z,3}, Q_x, Q_y)^T$,$\mathbb{A} = (A_x, A_y)$

$$A_x = \begin{pmatrix} 0 & 0 & -1/\mu_r \\ 0 & 0 & 0 \\ -1/\epsilon_{r,\infty} & 0 & 0 \end{pmatrix}, \quad A_y = \begin{pmatrix} 0 & 1/\mu_r & 0 \\ 1/\epsilon_{r,\infty} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

源项 $\mathbf{S}^{(1)}$ 为:

$$\mathbf{S}^{(1)} = \begin{pmatrix} -\left[\sigma + \epsilon_{r,\infty}(\sigma_x + \sigma_y)\right] E_z - \frac{\partial P}{\partial t} - (\sigma_x + \sigma_y)P + P_{z,1} + P_{z,2} + P_{z,3} \\ -\mu_r (\sigma_y - \sigma_x) H_x + Q_x \\ -\mu_r (\sigma_x - \sigma_y) H_y + Q_y \end{pmatrix},$$

对方程(5.1) 两边同乘 E_z，得到:

$$E_z \frac{\partial \mathbf{U}^{(1)}}{\partial t} + E_z \nabla \cdot (\mathbb{A}\mathbf{U}^{(1)}) = E_z \mathbf{S}^{(1)},$$

采用求解一维 Riemann 问题得到的数值通量 $\mathbf{h}_D\left(\mathbf{U}_N^{(1),-}, \mathbf{U}_N^{(1),+}\right)$ ，上式在积分下的表示为:

$$\begin{aligned} &\int_{\mathrm{D}} \left(\frac{\partial \mathbf{U}^{(1)}}{\partial t} E_z - \mathbf{S}^{(1)} E_z - \mathbb{A}\mathbf{U}^{(1)} \cdot \nabla E_z \right) d\mathbf{x} \\ &+ \int_{\partial \mathrm{D}} \mathbf{h}_D \left(\mathbf{U}^{(1),-}, \mathbf{U}^{(1),+} \right) E_z ds = 0, \end{aligned} \tag{5.2}$$

只需考虑上式中的第一分量，不将数值通量 $\mathbf{h}_D$ 展开，$\mathbf{S}^{E_z}$ 为源项的第一分量:

$$\int_D \left(\epsilon_{r,\infty} E_z \frac{\partial E_z}{\partial t} - \mathbf{S}^{E_z} E_z + H_y \frac{\partial E_z}{\partial x} - H_x \frac{\partial E_z}{\partial y} \right) dx + \int_{\partial D} \mathbf{h}_{D,1} E_z ds = 0, \tag{5.3}$$

同样的，将方程(5.1)两边同乘 H_x, H_y，并取对应的第二分量，第三分量，得到：

$$\begin{aligned}\int_D \left(\mu_r H_x \frac{\partial H_x}{\partial t} - \mathbf{S}^{H_x} H_x - E_z \frac{\partial H_x}{\partial y}\right) dx + \int_{\partial D} \mathbf{h}_{D,2} H_x ds = 0,\\ \int_D \left(\mu_r H_y \frac{\partial H_y}{\partial t} - \mathbf{S}^{H_y} H_y + E_z \frac{\partial H_y}{\partial x}\right) dx + \int_{\partial D} \mathbf{h}_{D,3} H_y ds = 0,\end{aligned} \tag{5.4}$$

先对数值通量不做展开，将 $\mathbf{S}^{E_z}, \mathbf{S}^{H_x}, \mathbf{S}^{H_y}$ 展开，得到：

$$\begin{aligned}&\int_D \left(\epsilon_{r,\infty} E_z \frac{\partial E_z}{\partial t} + H_y \frac{\partial E_z}{\partial x} - H_x \frac{\partial E_z}{\partial y}\right) dx + \int_{\partial D} \mathbf{h}_{D,1} E_z ds = \\ &\int_D \left(-\left[\sigma + \epsilon_{r,\infty}\left(\sigma_x + \sigma_y\right)\right] E_z^2 - \frac{\partial P}{\partial t} E_z - (\sigma_x + \sigma_y) P E_z + P_{z,1} E_z + P_{z,2} E_z + P_{z,3} E_z\right) dx,\end{aligned} \tag{5.5}$$

E_z part

$$\int_D \left(\mu_r H_x \frac{\partial H_x}{\partial t} - E_z \frac{\partial H_x}{\partial y}\right) dx + \int_{\partial D} \mathbf{h}_{D,2} H_x ds = \int_D \left(-\mu_r \left(\sigma_y - \sigma_x\right) H_x^2 + Q_x H_x\right) dx, \tag{5.6}$$

H_x part

$$\int_D \left(\mu_r H_y \frac{\partial H_y}{\partial t} + E_z \frac{\partial H_y}{\partial x}\right) dx + \int_{\partial D} \mathbf{h}_{D,3} H_y ds = \int_D \left(-\mu_r \left(\sigma_x - \sigma_y\right) H_y^2 + Q_y H_y\right) dx, \tag{5.7}$$

H_y part

对于广义状态方程部分，同样两端乘上相应的项，得到如下的结果：

$$P \frac{\partial P}{\partial t} + \tau^\alpha \frac{\partial^\alpha P}{\partial t^\alpha} \frac{\partial P}{\partial t} = (\epsilon_{r,s} - \epsilon_{r,\infty}) E_z \frac{\partial P}{\partial t}, \tag{5.8a}$$

P part

$$P_{z,1} \frac{\partial P_{z,1}}{\partial t} = -\epsilon_{r,\infty} \sigma_x \sigma_y E_z P_{z,1}, \tag{5.8b}$$

P_{z,1} part

$$P_{z,2} \frac{\partial P_{z,2}}{\partial t} = -\sigma_x \sigma_y P P_{z,2}, \tag{5.8c}$$

P_{z,2} part

$$P_{z,3} \frac{\partial P_{z,3}}{\partial t} = \frac{\sigma}{\epsilon_{r,\infty}} P_{z,1} P_{z,3} - \sigma(\sigma_x + \sigma_y) E_z P_{z,3}, \tag{5.8d}$$

P_{z,3} part

$$Q_x \frac{\partial Q_x}{\partial t} = -\sigma_x Q_x^2 + \mu_r \sigma_x \left(\sigma_y - \sigma_x\right) H_x Q_x, \tag{5.8e}$$

Q_x part

$$Q_y \frac{\partial Q_y}{\partial t} = -\sigma_y Q_y^2 + \mu_r \sigma_y \left(\sigma_x - \sigma_y\right) H_y Q_y, \tag{5.8f}$$

Q_y part

基于第3节所分析的一维情况下对于 Cole-Cole 介质中带有 UPML 吸收层的 Maxwell 方程组的稳定性，我们研究其在二维情况下，能量函数应该如何给出。

5.1 usual 2D rectangular area

目前对 2 维矩形计算区域设计其四边的 UPML 吸收边界层。

仍然先给出频域下，带有 UPML 吸收边界层的二维 Maxwell 方程组的形式，在做分数阶 Maxwell 相关的论文 [10, 13, 14] 中都没有考虑 $J = \sigma E$ 电流强度项，暂时也不做考虑：

$$\frac{\partial \breve{H}_y}{\partial x} - \frac{\partial \breve{H}_x}{\partial y} = \left(j\omega\epsilon_0\epsilon_{r,\infty}\breve{E}_z + j\omega\breve{P}\right) s_x s_y, \tag{5.9a}$$

$$\frac{\partial \breve{E}_z}{\partial y} = -j\omega\mu_0\mu_r \frac{s_y}{s_x}\breve{H}_x, \tag{5.9b}$$

$$-\frac{\partial \breve{E}_z}{\partial x} = -j\omega\mu_0\mu_r \frac{s_x}{s_y}\breve{H}_y, \tag{5.9c}$$

$$\breve{P} + (j\omega\tau)^{\alpha}\breve{P} = \epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty})\breve{E}_z. \tag{5.9d}$$

其中 $s_i = 1 + \frac{\sigma_i}{j\omega\epsilon_0}, \quad i = x, y$ 是有关 UPML 层的取值，具体的取值方法在 section 2中提及。

化简后的方程组为：

$$\epsilon_0\epsilon_{r,\infty}\frac{\partial E_z}{\partial t} = \frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} - \epsilon_{r,\infty}(\sigma_x + \sigma_y)E_z - \frac{\partial P}{\partial t} - \frac{\sigma_x + \sigma_y}{\epsilon_0}P + P_{z,1} + P_{z,2}, \tag{5.10a}$$

$$\mu_0\mu_r\frac{\partial H_x}{\partial t} = -\frac{\partial E_z}{\partial y} - \mu_0\mu_r\frac{\sigma_y - \sigma_x}{\epsilon_0}H_x + Q_x, \tag{5.10b}$$

$$\mu_0\mu_r\frac{\partial H_y}{\partial t} = \frac{\partial E_z}{\partial x} - \mu_0\mu_r\frac{\sigma_x - \sigma_y}{\epsilon_0}H_y + Q_y, \tag{5.10c}$$

$$P + \tau^{\alpha}\frac{\partial^{\alpha} P}{\partial t^{\alpha}} = \epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty})E_z, \tag{5.10d}$$

$$\frac{\partial P_{z,1}}{\partial t} = -\frac{\epsilon_{r,\infty}}{\epsilon_0}\sigma_x\sigma_y E_z, \tag{5.10e}$$

$$\frac{\partial P_{z,2}}{\partial t} = -\frac{\sigma_x\sigma_y}{\epsilon_0^2}P, \tag{5.10f}$$

$$\frac{\partial Q_x}{\partial t} = -\frac{\sigma_x}{\epsilon_0}Q_x + \frac{\mu_0\mu_r\sigma_x\left(\sigma_y - \sigma_x\right)}{\epsilon_0^2}H_x, \tag{5.10g}$$

$$\frac{\partial Q_y}{\partial t} = -\frac{\sigma_y}{\epsilon_0}Q_y + \frac{\mu_0\mu_r\sigma_y\left(\sigma_x - \sigma_y\right)}{\epsilon_0^2}H_y, \tag{5.10h}$$

边界条件取做 perfectly conducting boundary condition:

$$\hat{\boldsymbol{n}} \times E = 0 \quad \text{on } \partial\Omega, \tag{5.11}$$

其中 $\hat{\boldsymbol{n}}$ 为 $\partial\Omega$ 上的外法向量，初值条件为：

$$\begin{aligned}
&E_z(x,y,0)=E_0(x,y),\quad P(x,y,0)=P_0(x,y),\\
&H_y(x,y,0)=H_{y,0}(x,y),\quad H_x(x,y,0)=H_{x,0}(x,y),\\
&P_{z,1}(x,y,0)=P_{z,1,0}(x,y),\quad P_{z,2}(x,y,0)=P_{z,2,0}(x,y),\\
&Q_x(x,y,0)=Q_{x,0}(x,y),\quad Q_y(x,y,0)=Q_{y,0}(x,y).
\end{aligned}\tag{5.12}$$

将 $\frac{\partial^\alpha P}{\partial t^\alpha}$ 用辅助变量进行表示：

$$\frac{\partial^\alpha P(x,t)}{\partial t^\alpha}=\int_0^\infty \varphi(x,t;\lambda)d\lambda,\tag{5.13}$$

其中辅助变量 φ 由下述 ODE 方程进行描述

$$\begin{aligned}
&\frac{\partial\varphi}{\partial t}=-\lambda\varphi+\frac{\sin(\pi\alpha)}{\pi}\lambda^{\alpha-1}\frac{\partial P}{\partial t},\quad \lambda>0,\\
&\varphi(x,0;\lambda)=0,\quad \lambda>0.
\end{aligned}\tag{5.14}$$

Auxiliary Equat

也可以将上述 ODE 的解直接写做：

$$\varphi(x,t;\lambda):=\frac{\sin(\pi\alpha)}{\pi}\lambda^{\alpha-1}\int_0^t e^{-(t-\tau)\lambda}\frac{\partial P(x,\tau)}{\partial\tau}d\tau.\tag{5.15}$$

5.1.1 2D rectangular energy function

对方程组(5.10)和辅助方程(5.14)构造能量函数：

$$W=\int_\Omega\left(\frac{\epsilon_0\epsilon_\infty}{2}E_z^2+\frac{a_1}{2}P^2+\frac{b_1}{2}P_{z,1}^2+\frac{b_2}{2}P_{z,2}^2+\frac{1}{2}\int_0^\infty b_3\varphi^2d\lambda+\frac{\mu_0\mu_r}{2}H_x^2+\frac{\mu_0\mu_r}{2}H_y^2+\frac{1}{2}Q_x^2+\frac{1}{2}Q_y^2\right)dS\tag{5.16}$$

其中 E_z 表示电场强度，P 表示电极化强度，H_x, H_y 为磁场强度，$P_{z,1}, P_{z,2}, \varphi, Q_x, Q_y$ 均为辅助变量。并且其中 a_1, b_1, b_2, b_3 均为标量。

显然这个能量函数 $W \geq 0$，将其对时间 t 做导数：

$$\begin{aligned}
\frac{dW}{dt} &= \int_\Omega \left(\epsilon_0 \epsilon_\infty E_z \frac{\partial E_z}{\partial t} + a_1 P \frac{\partial P}{\partial t} + b_1 P_{z,1} \frac{\partial P_{z,1}}{\partial t} + b_2 P_{z,2} \frac{\partial P_{z,2}}{\partial t} + \int_0^\infty b_3 \varphi \frac{\partial \varphi}{\partial t} d\lambda \right. \\
&\quad \left. + \mu_0 \mu_r (H_x \frac{\partial H_x}{\partial t} + H_y \frac{\partial H_y}{\partial t}) + Q_x \frac{\partial Q_x}{\partial t} + Q_y \frac{\partial Q_y}{\partial t} \right) dS \\
&= \int_\Omega \left(E_z \frac{\partial H_y}{\partial x} + H_y \frac{\partial E_z}{\partial x} - E_z \frac{\partial H_x}{\partial y} - H_x \frac{\partial E_z}{\partial y} - \epsilon_{r,\infty} (\sigma_x + \sigma_y) E_z^2 + \left[a_1 - \frac{1}{\epsilon_0 (\epsilon_{r,s} - \epsilon_{r,\infty})} \right] P \frac{\partial P}{\partial t} \right. \\
&\quad - \frac{\tau^\alpha}{\epsilon_0 (\epsilon_{r,s} - \epsilon_{r,\infty})} \int_0^\infty \left(\frac{\pi}{\sin \pi\alpha} \lambda^{1-\alpha} \varphi \frac{\partial \varphi}{\partial t} + \frac{\pi}{\sin \pi\alpha} \lambda^{2-\alpha} \varphi^2 \right) d\lambda - \frac{\sigma_x + \sigma_y}{\epsilon_0} E_z P \\
&\quad + (1 - b_1 \frac{\epsilon_{r,\infty}}{\epsilon_0}) E_z P_{z,1} + E_z P_{z,2} - b_2 \frac{\sigma_x \sigma_y}{\epsilon_0^2} P P_{z,2} + \int_0^\infty b_3 \varphi \frac{\partial \varphi}{\partial t} d\lambda - \mu_r (\sigma_y - \sigma_x) H_x^2 \\
&\quad \left. + \left[1 + \mu_r \sigma_x (\sigma_y - \sigma_x) \right] H_x Q_x - \sigma_x Q_x^2 - \mu_r (\sigma_x - \sigma_y) H_y^2 + \left[1 + \mu_r \sigma_y (\sigma_x - \sigma_y) \right] H_y Q_y - \sigma_y Q_y^2 \right) dS
\end{aligned} \tag{5.17}$$

dW /dt in 2D re

可以证明得到

Theorem 5.1. 能量函数 W 是有界的，*i.e.* $W(t) \leq CW(0)$ 对任意 $t \in [0, T]$ 成立，其中常数 C 依赖于 T.

Remark 5.1. 这个定理说明了能量函数 W 是 *well-posed*, 但是对于 *Maxwell's* 方程组而言，其解 $u(t)$ 并不能说是 *stable* 的。*stable* 的定义是指上界 $C(t) = 1 + t^p$，也就是被一个多项式限制住

证明. 与 [9] 中分析的结果一致，只需证明 $\frac{dW}{dt} \leq cW$，其中 $c > 0$，再利用 Gronwall 不等式得到：

$$W(t) \leq e^{ct} W(0) \leq e^{cT} W(0)$$

对于空间导数项，利用完全导电边界条件 (PEC) ，E_z 在边界上的取值为 0，得到：

$$\int_\Omega \left(E_z \frac{\partial H_y}{\partial x} + H_y \frac{\partial E_z}{\partial x} - E_z \frac{\partial H_x}{\partial y} - H_x \frac{\partial E_z}{\partial y} \right) dS = 0. \tag{5.18}$$

对于 Q_x, H_x, Q_y, H_y 的项，将 $\frac{dW}{dt} \leq cW$ 的相关项整理完成之后，得到：

$$\begin{aligned}
&\left[\frac{c}{2} \mu_0 \mu_r + \mu_r (\sigma_y - \sigma_x) \right] H_x^2 - \left[1 + \mu_r \sigma_x (\sigma_y - \sigma_x) \right] H_x Q_x + \left(\frac{c}{2} + \sigma_x \right) Q_x^2 \\
+ &\left[\frac{c}{2} \mu_0 \mu_r + \mu_r (\sigma_x - \sigma_y) \right] H_y^2 - \left[1 + \mu_r \sigma_y (\sigma_x - \sigma_y) \right] H_y Q_y + \left(\frac{c}{2} + \sigma_y \right) Q_y^2 \geq 0
\end{aligned} \tag{5.19}$$

关于 Q_x^2, Q_y^2 项的系数大于 0，需要考虑上述不等式关于 Q_x, Q_y 的判别式：

$$\begin{aligned}\Delta_1 &= [1+\mu_r\sigma_x(\sigma_y-\sigma_x)]^2 - 4\left[\frac{c}{2}\mu_0\mu_r + \mu_r(\sigma_y-\sigma_x)\right]\left(\frac{c}{2}+\sigma_x\right) \le 0\\ \Delta_2 &= [1+\mu_r\sigma_y(\sigma_x-\sigma_y)]^2 - 4\left[\frac{c}{2}\mu_0\mu_r + \mu_r(\sigma_x-\sigma_y)\right]\left(\frac{c}{2}+\sigma_y\right) \le 0\end{aligned} \tag{5.20}$$

由章节2中 σ_x, σ_y 的取法可知 σ_x, σ_y 都是有限值，且 $c>0$，则当

$$c \ge \max\{\frac{4\sigma_{\max}}{\epsilon_0}, \frac{3\left(1+2\frac{\mu_0\mu_r}{\epsilon_0^2}\sigma_{\max}^2\right)^2}{\sqrt{\mu_0\mu_r}}\}$$

时，$\Delta_{1,2} \le 0$ 成立。

对于 $E_z, P, P_{z,1}, P_{z,2}, \varphi$ 项，将 $\frac{dW}{dt} \le cW$ 的相关项整理完成之后取 $a_1 = \frac{1}{\epsilon_0(\epsilon_{r,s}-\epsilon_{r,\infty})}$，$b_3 = \frac{\tau_0^\alpha}{\epsilon_0(\epsilon_{r,s}-\epsilon_{r,\infty})}\frac{\pi}{\sin\pi\alpha}\lambda^{1-\alpha}$，得到：

$$\begin{aligned}&\left[\frac{c}{2}\epsilon_0\epsilon_{r,\infty} + \epsilon_{r,\infty}(\sigma_x+\sigma_y)\right]E_z^2 + \frac{c}{2}a_1P^2 + \frac{c}{2}b_1P_{z,1}^2 + \frac{c}{2}b_2P_{z,2}^2 + \frac{\sigma_x+\sigma_y}{\epsilon_0}E_zP\\ &-\left(1-b_1\frac{\epsilon_{r,\infty}}{\epsilon_0}\sigma_x\sigma_y\right)E_zP_{z,1} - E_zP_{z,2} + b_2\frac{\sigma_x\sigma_y}{\epsilon_0^2}PP_{z,2}\\ &+\int_0^\infty\left(\frac{cb_3}{2} + \frac{\tau^\alpha}{\epsilon_0(\epsilon_{r,s}-\epsilon_{r,\infty})}\frac{\pi}{\sin\pi\alpha}\lambda^{2-\alpha}\right)\varphi^2 d\lambda \ge 0,\end{aligned} \tag{5.21}$$

注意到能量函数的取法，$a_1, b_1, b_2, b_3 \ge 0$，则关于 φ^2 的系数大于 0，对于剩下的项，整理为：

$$\begin{aligned}&\epsilon_{r,\infty}(\sigma_x+\sigma_y)E_z^2 + \left[\frac{c}{6}\epsilon_0\epsilon_{r,\infty}E_z^2 + \frac{\sigma_x+\sigma_y}{\epsilon_0}E_zP + \frac{c}{4}a_1P^2\right]\\ &+\left[\frac{c}{6}\epsilon_0\epsilon_{r,\infty}E_z^2 - \left(1-b_1\frac{\epsilon_{r,\infty}}{\epsilon_0}\sigma_x\sigma_y\right)E_zP_{z,1} + \frac{c}{2}b_1P_{z,1}^2\right]\\ &+\left[\frac{c}{6}\epsilon_0\epsilon_{r,\infty}E_z^2 - E_zP_{z,2} + \frac{c}{4}b_2P_{z,2}^2\right] + \left[\frac{c}{4}a_1P^2 + b_2\frac{\sigma_x\sigma_y}{\epsilon_0^2}PP_{z,2} + \frac{c}{4}b_2P_{z,2}^2\right] \ge 0,\end{aligned} \tag{5.22}$$

同样需要考虑其判别式：

$$\begin{aligned}\Delta_3 &= \left(\frac{\sigma_x+\sigma_y}{\epsilon_0}\right)^2 - \frac{c^2}{6}\epsilon_0\epsilon_{r,\infty}a_1 \le 0,\\ \Delta_4 &= \left(1-b_1\frac{\epsilon_{r,\infty}}{\epsilon_0}\sigma_x\sigma_y\right)^2 - \frac{c^2}{3}\epsilon_0\epsilon_{r,\infty}b_1 \le 0,\\ \Delta_5 &= 1 - \frac{c^2}{6}\epsilon_0\epsilon_{r,\infty}b_2 \le 0,\\ \Delta_6 &= b_2^2\left(\frac{\sigma_x\sigma_y}{\epsilon_0^2}\right)^2 - \frac{c^2}{4}a_1b_2 \le 0,\end{aligned} \tag{5.23}$$

同样由于σ_x, σ_y的取值是有限值，则当

$$c \geq \max\{c_3, c_4, c_5, c_6\}$$

时，$\Delta_{3,4,5,6} \leq 0$成立。那么取$c \geq \max\{c_1, c_2, c_3, c_4, c_5, c_6\}$时，有$\frac{dW}{dt} \leq cW$成立，得到定理中的结果。 □

Remark 5.2. 这样的能量函数W与论文 *[9]* 中的证明结果匹配，但只能满足在有限时间T内是有限的，但其不是递减的函数，甚至有可能存在某个时刻t，$W(t) \geq W(0)$成立。

5.2 2D rectangular area with UPML in x-axis

对2维矩形计算区域只考虑在x方向上匹配上吸收边界层，使在吸收边界层中不出现现corner区域。如图1所示

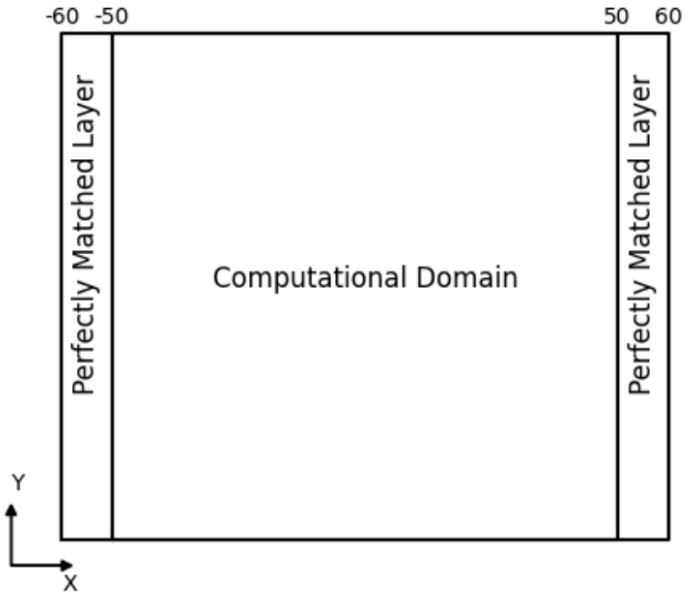


图 1: 在x轴方向上的吸收边界层 [1]

UPML_x_axis

对应为方程组(5.9a)-(5.9d)中 $s_y = 1$，即 $\sigma_y = 0$ 的情形。即：

$$\begin{aligned}
\frac{\partial \breve{H}_y}{\partial x} - \frac{\partial \breve{H}_x}{\partial y} &= \left(j\omega\epsilon_0\epsilon_{r,\infty}\breve{E}_z + j\omega\breve{P}\right) s_x, \\
\frac{\partial \breve{E}_z}{\partial y} &= -j\omega\mu_0\mu_r \frac{1}{s_x}\breve{H}_x, \\
-\frac{\partial \breve{E}_z}{\partial x} &= -j\omega\mu_0\mu_r s_x \breve{H}_y, \\
\breve{P} + (j\omega\tau)^\alpha \breve{P} &= \epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty})\breve{E}_z.
\end{aligned} \tag{5.24}$$

frequency Maxwe

将其化简至时域中，即为：

$$\epsilon_0\epsilon_{r,\infty}\frac{\partial E_z}{\partial t} = \frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} - \epsilon_{r,\infty}\sigma_x E_z - \frac{\partial P}{\partial t} - \frac{\sigma_x}{\epsilon_0}P, \tag{5.25a}$$

2D maxwell x-ax

$$P + \tau^\alpha \frac{\partial^\alpha P}{\partial t^\alpha} = \epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty})E_z, \tag{5.25b}$$

2D maxwell x-ax

$$\mu_0\mu_r\frac{\partial H_x}{\partial t} = -\frac{\partial E_z}{\partial y} + \mu_0\mu_r\frac{\sigma_x}{\epsilon_0}H_x + Q_x, \tag{5.25c}$$

2D maxwell x-ax

$$\mu_0\mu_r\frac{\partial H_y}{\partial t} = \frac{\partial E_z}{\partial x} - \mu_0\mu_r\frac{\sigma_x}{\epsilon_0}H_y, \tag{5.25d}$$

2D maxwell x-ax

$$\frac{\partial Q_x}{\partial t} = -\frac{\sigma_x}{\epsilon_0}Q_x - \mu_0\mu_r\frac{\sigma_x^2}{\epsilon_0^2}H_x \tag{5.25e}$$

2D maxwell x-ax

2D with s_y = 1

期望找到一个能量函数 $W(t)$ 使得其随着时间 t 递减，或至少满足 $W(t) \le W(0)$。

假设 2 维 TM 平面波入射到界面，入射面是 xOy，磁场位于 (x, y) 平面上，电场只有垂直于该平面的分量，即 $H = H_x(x, y, t)e_x + H_y(x, y, t)e_y, \quad E = E_z(x, y, t)e_z$

假设电磁波在 xOy 平面中沿着一个特定的方向传播，对于任意一种场变量 F，假定其具备形式：

$$F(x, y, t) = \widetilde{F}(t)e^{-jk_x x - jk_y y}$$

将 y 方向上的依赖性固定用 $e^{-jk_y y}$ 表示，对 x 方向上的依赖性固定用 $e^{-jk_x x}$ 表示。在 y 方向上是均匀的，并且具有周期性或无限长度的系统。

在进行计算时，需要考虑所有方向上波的解，对于每一个 k_x, k_y，都存在一个 $-k_y$ 与之

对应，所以对于2维TM波中，各个变量在频率下设置为：

$$\begin{aligned}
\breve{H}_x(x,y,\omega) =& \breve{H}_{x0}(\omega)e^{-jk_x x-jk_y y}\\
\breve{H}_y(x,y,\omega) =& \breve{H}_{y0}(\omega)e^{-jk_x x-jk_y y}\\
\breve{E}_z(x,y,\omega) =& \breve{E}_{z0}(\omega)e^{-jk_x x-jk_y y}\\
\breve{P}(x,y,\omega) =& \breve{P}_0(\omega)e^{-jk_x x-jk_y y}\\
\breve{Q}(x,y,\omega) =& \breve{Q}_0(\omega)e^{-jk_x x-jk_y y}
\end{aligned} \tag{5.26}$$

带入方程组(5.24)中化简得到：

$$\epsilon_0\epsilon_{r,\infty}\frac{\partial E_{z0}}{\partial t} = -jk_x H_{y0} + jk_y H_{x0} - \epsilon_{r,\infty}\sigma_x E_{z0} - \frac{\partial P_0}{\partial t} - \frac{\sigma_x}{\epsilon_0}P_0, \tag{5.27a}$$

$$P_0 + \tau^\alpha\frac{\partial^\alpha P_0}{\partial t^\alpha} = \epsilon_0(\epsilon_{r,s}-\epsilon_{r,\infty})E_{z0}, \tag{5.27b}$$

$$\mu_0\mu_r\frac{\partial H_{x0}}{\partial t} = jk_y E_{z0} + \mu_0\mu_r\frac{\sigma_x}{\epsilon_0}H_{x0} + Q_{x0}, \tag{5.27c}$$

$$\mu_0\mu_r\frac{\partial H_{y0}}{\partial t} = -jk_x E_{z0} - \mu_0\mu_r\frac{\sigma_x}{\epsilon_0}H_{y0}, \tag{5.27d}$$

$$\frac{\partial Q_{x0}}{\partial t} = -\frac{\sigma_x}{\epsilon_0}Q_{x0} - \mu_0\mu_r\frac{\sigma_x^2}{\epsilon_0^2}H_{x0}, \tag{5.27e}$$

wave direction

与方程组(5.25)不同的是，原方程中变量H_x, H_y, E_z, P, Q_x均为实数，但在方程组(5.27)中，变量$H_{x0}, H_{y0}, E_{z0}, P_0, Q_{x0}$均为复数。

将能量函数取做

$$\begin{aligned}
W = \int_\Omega \Big[&a_1\mu_0\mu_r H_{x0}^* H_{x0} + a_2\mu_0\mu_r H_{y0}^* H_{y0} + a_3 Q_{x0}^* Q_{x0} + b_1\mu_0\mu_r\left(Q_{x0}^* H_{x0} + H_{x0}^* Q_{x0}\right)\\
&+ b_2\mu_0\mu_r\left(H_{x0}^* H_{y0} + H_{y0}^* H_{x0}\right) + b_3\mu_0\mu_r\left(Q_{x0}^* H_{y0} + H_{y0}^* Q_{x0}\right)\\
&+a_4\epsilon_0\epsilon_{r,\infty}E_{z0}^* E_{z0} + a_5 P_0^* P_0 + \int_0^\infty a_6\varphi_0^*\varphi_0 d\lambda + \int_0^\infty b_4\left(\varphi_0^* P_0 + P_0^*\varphi_0\right)d\lambda\Big]\, dx
\end{aligned} \tag{5.28}$$

但对于这类型的能量函数，需要其满足$W \geq 0, dW/dt \leq 0$，其中电场强度E_{z0}与电极化强度P_0和一维中处理的方法相同，剩下对磁场强度H_{x0}, H_{y0}，以及相应的辅助变量Q_{x0}满足的条件等价于

$$\begin{pmatrix} a_1 & b_2 & b_1 \\ b_2 & a_2 & b_3 \\ b_1 & b_3 & \dfrac{a_3}{\mu_0\mu_r} \end{pmatrix} \geq 0 \tag{5.29}$$

$$\begin{pmatrix} 2a_4\epsilon_{r,\infty}\sigma_x & a_1jk_y - b_2jk_x - a_4jk_y & -(a_2jk_x - b_2jk_y - a_4jk_x) & b_1jk_y - b_3jk_x \\ -(a_1jk_y - b_2jk_x - a_4jk_y) & 2\mu_0\mu_r\frac{\sigma_x}{\epsilon_0}\left(b_1\mu_0\mu_r\frac{\sigma_x}{\epsilon_0} - a_1\right) & b_3\mu_0^2\mu_r^2\frac{\sigma_x^2}{\epsilon_0^2} & a_3\mu_0\mu_r\frac{\sigma_x^2}{\epsilon_0^2} - a_1 \\ a_2jk_x - b_2jk_y - a_4jk_x & b_3\mu_0^2\mu_r^2\frac{\sigma_x^2}{\epsilon_0^2} & 2a_2\mu_0\mu_r\frac{\sigma_x}{\epsilon_0} & 2b_3\mu_0\mu_r\frac{\sigma_x}{\epsilon_0} - b_2 \\ -(b_1jk_y - b_3jk_x) & a_3\mu_0\mu_r\frac{\sigma_x^2}{\epsilon_0^2} - a_1 & 2b_3\mu_0\mu_r\frac{\sigma_x}{\epsilon_0} - b_2 & 2a_3\frac{\sigma_x}{\epsilon_0} - 2b_1 \end{pmatrix} \geq 0 \tag{5.30}$$

一般的，可以取 $\mu_0 = \mu_r = \epsilon_0 = \epsilon_{r,\infty} = 1$ 相当于无量纲化后的结果。再不妨取 $a_4 = 1$，上述的两种不等式条件在数值上不成立。

在数值上的取法为：$a_i = 0:1:10, b_i = 0:1:10, \sigma_x = 0:0.1:10, \quad i = 1,2,3$

5.3 2D rectangular area with stretched coordinate PML in x-axis

考虑stretched coordinate PML, 在MIT的课程note：Coordinate Transformation Invariance in Electromagnetism中证明到了stretched coordinate PML的表达形式和UPML在形式上具备相似性。

取非奇异的变换 $\mathbf{x} \to \mathbf{x}'$，用 $\mathcal{J}$ 表示对应的Jacobian矩阵，即

$$\mathcal{J}_{ij} = \frac{\partial x_i'}{\partial x_j} \tag{5.31}$$

利用链式法则，对应的变量在新的坐标下表示为：

$$\begin{aligned} \mathbf{E}' &= \left(\mathcal{J}^T\right)^{-1}\mathbf{E}, \\ \mathbf{H}' &= \left(\mathcal{J}^T\right)^{-1}\mathbf{H}, \\ \boldsymbol{\epsilon}' &= \frac{\mathcal{J}\boldsymbol{\epsilon}\mathcal{J}^T}{\det\mathcal{J}}, \\ \boldsymbol{\mu}' &= \frac{\mathcal{J}\boldsymbol{\mu}\mathcal{J}^T}{\det\mathcal{J}}, \end{aligned} \tag{5.32}$$

特别的，尽管在坐标变换前是各向同性的介质（$\boldsymbol{\epsilon}, \boldsymbol{\mu}$ 是标量），但在坐标变换后通常会获得各向异性的介质（$\boldsymbol{\epsilon}, \boldsymbol{\mu}$ 是张量）。

对于原Maxwell's Equation

$$\begin{aligned} \nabla\times\mathbf{H} &= \boldsymbol{\epsilon}\frac{\partial\mathbf{E}}{\partial t}, \\ \nabla\times\mathbf{E} &= -\boldsymbol{\mu}\frac{\partial\mathbf{H}}{\partial t}, \end{aligned} \tag{5.33}$$

转换为

$$\begin{aligned}\nabla' \times \mathbf{H}' &= \boldsymbol{\epsilon}' \frac{\partial \mathbf{E}'}{\partial t} = \frac{\mathcal{J}\boldsymbol{\epsilon}\mathcal{J}^T}{\det \mathcal{J}} \frac{\partial \mathbf{E}'}{\partial t}, \\ \nabla' \times \mathbf{E}' &= -\boldsymbol{\mu}' \frac{\partial \mathbf{H}'}{\partial t} = -\frac{\mathcal{J}\boldsymbol{\mu}\mathcal{J}^T}{\det \mathcal{J}} \frac{\partial \mathbf{H}'}{\partial t},\end{aligned} \tag{5.34}$$

Maxwell's Equat

当取

$$\frac{\partial}{\partial x} \to \frac{1}{s_x} \frac{\partial}{\partial x'} \tag{5.35}$$

Specifically, for a PML designed to absorb waves propagating in the x direction, the following transformation is included in the wave equation. Wherever an x derivative $\partial/\partial x$ appears in the wave equation, it is replaced by:

$$\frac{\partial}{\partial x} \to \frac{1}{1 + \frac{i\sigma(x)}{\omega}} \frac{\partial}{\partial x} \tag{5.36}$$

$$\mathcal{J} = \begin{bmatrix} \frac{1}{s_x} & 0 & 0 \\ 0 & \frac{1}{s_y} & 0 \\ 0 & 0 & \frac{1}{s_z} \end{bmatrix}, \boldsymbol{\epsilon} = \epsilon_a \mathcal{I}, \boldsymbol{\mu} = \mu_a \mathcal{I} \tag{5.37}$$

时，stretched coordinate PML 的表达形式和 UPML 在形式上具备相似性。都是在 Maxwell's Equations 方程的基础上，空间导数项和时间导数项保持不变，对介电系数 ϵ 和磁导系数 μ 进行变换。

$$\frac{\mathcal{J}\mathcal{J}^T}{\det \mathcal{J}} = \begin{bmatrix} \frac{s_y s_z}{s_x} & 0 & 0 \\ 0 & \frac{s_x s_z}{s_y} & 0 \\ 0 & 0 & \frac{s_x s_y}{s_z} \end{bmatrix} = \bar{\bar{\epsilon}} = \bar{\bar{\mu}} \tag{5.38}$$

注意到，频率下 1 维 Cole-Cole 介质中 Maxwell 方程组表达为

$$\begin{aligned}\frac{\partial \hat{H}}{\partial x} &= j\omega \left(\varepsilon_0 \varepsilon_\infty \hat{E} + \hat{P} \right) \\ \frac{\partial \hat{E}}{\partial x} &= j\omega \mu_0 \hat{H}\end{aligned} \tag{5.39}$$

其中电极化强度 P 和电场强度 E 的关系和方程(3.1)中的表达式一致

在 UPML 的吸收边界下，1 维 Maxwell's Equations 变为

$$\begin{aligned}\frac{\partial \hat{H}_y}{\partial x} &= j\omega \left(\varepsilon_0 \varepsilon_\infty \hat{E}_z + \hat{P}_z \right) s_x \\ \frac{\partial \hat{E}_z}{\partial x} &= j\omega \mu_0 \hat{H}_y s_x\end{aligned} \tag{5.40}$$

而在 stretched coordinate PML 下，1 维 Maxwell's Equations 变为

$$\begin{aligned}\frac{\partial \hat{H}'}{\partial x'} &= j\omega\left(\varepsilon_0\varepsilon_\infty \hat{E}' + \hat{P}'\right) s_x \\ \frac{\partial \hat{E}'}{\partial x'} &= j\omega\mu_0 \hat{H}' s_x\end{aligned} \tag{5.41}$$

故此这两种吸收边界层的表达形式在形式上具备相似性。在之后我们均采用 stretched coordinate PML 进行描述。

在二维的特殊情况中，即 $\sigma_x \neq 0, \sigma_y = 0$，假定某个波的传播方向为 (k_x, k_y)，如图所示

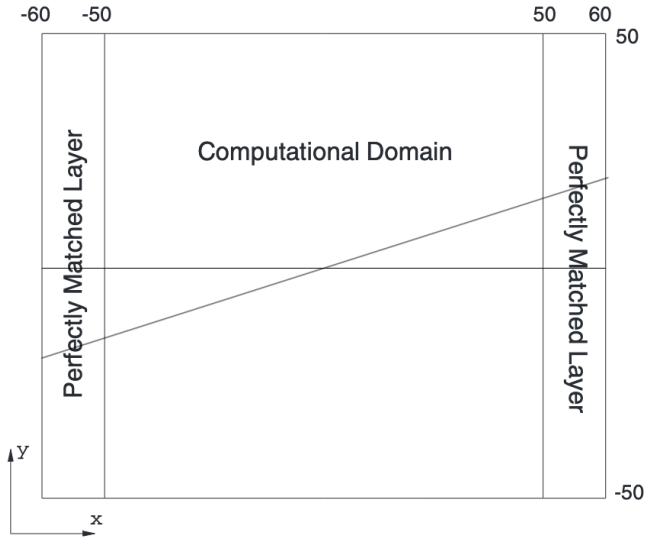


图 2: 不同传播方向在 x 轴方向上的吸收边界层 [1]

UPML_x_axis_2

定义 3 维情形下的旋转算子 $\mathcal{R}(\theta)$

$$\mathcal{R}(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{5.42}$$

其中 $\tan\theta = k_y/k_x$

表达在 3 维情形下，stretched coordinate PML 中对应的将任意一个倾斜方向转换至水

平 x 轴方向的 transformer matrix 为

$$\mathcal{Q} = \mathcal{J}\mathcal{R}(-\theta) \quad \text{or} \quad \mathcal{R}(-\theta)\mathcal{J} \tag{5.43}$$

在这个坐标变换下，仍然取介电常数参数及磁导率参数为 $\epsilon_a\mathcal{I}, \mu_a\mathcal{I}$，则在 $\mathcal{Q}$ 坐标变换下的 Maxwell's Equations 为

$$\begin{aligned} \nabla' \times \mathbf{H}' &= \frac{\mathcal{Q}\boldsymbol{\epsilon}\mathcal{Q}^T}{\det \mathcal{Q}} \frac{\partial \mathbf{E}'}{\partial t}, \\ \nabla' \times \mathbf{E}' &= -\frac{\mathcal{Q}\boldsymbol{\mu}\mathcal{Q}^T}{\det \mathcal{Q}} \frac{\partial \mathbf{H}'}{\partial t}, \end{aligned} \tag{5.44}$$

Maxwell's Equat

若是先旋转再伸缩，即 $\mathcal{Q} = \mathcal{J}\mathcal{R}(-\theta)$，那么

$$\frac{\mathcal{Q}\boldsymbol{\epsilon}\mathcal{Q}^T}{\det \mathcal{Q}} = \frac{\mathcal{J}\mathcal{R}(-\theta)\epsilon_a\mathcal{I}\mathcal{R}(-\theta)^T\mathcal{J}^T}{\det\left(\mathcal{J}\mathcal{R}(-\theta)\right)} = \epsilon_a \frac{\mathcal{J}\mathcal{J}^T}{\det \mathcal{J}} \tag{5.45}$$

但若是先伸缩再旋转，即 $\mathcal{Q} = \mathcal{R}(-\theta)\mathcal{J}$，那么

$$\frac{\mathcal{Q}\boldsymbol{\epsilon}\mathcal{Q}^T}{\det \mathcal{Q}} = \frac{\mathcal{R}(-\theta)\mathcal{J}\epsilon_a\mathcal{I}\mathcal{J}^T\mathcal{R}(-\theta)^T}{\det\left(\mathcal{R}(-\theta)\mathcal{J}\right)} = \epsilon_a \frac{\mathcal{R}(-\theta)\mathcal{J}\mathcal{J}^T\mathcal{R}(-\theta)^T}{\det \mathcal{J}} \tag{5.46}$$

对于算子 $\mathcal{Q}$ 是先旋转再伸缩，还是先伸缩再旋转，会导致作用后的 Maxwell's equations 具有不同的形式，两者相等当且仅当 $\mathcal{J}\mathcal{J}^T = \Lambda\mathcal{I}$ 时成立。

作用完算子 $\mathcal{Q} = \mathcal{J}\mathcal{R}(-\theta)$ 后，(k_x, k_y) 方向的波已经旋转到了 x 轴方向，即 $k_y' = 0$，则只用考虑方程组(5.44)中关于 x 方向的方程即可。这样与一维情形保持一致。

需要说明的是在旋转后的吸收层参数 σ_x 和一维中的表达可能不一样。一维中的吸收参数 σ_x 是在 x 方向上的，而在二维中，旋转后的吸收参数 σ_x 是在旋转后的 x 方向上的，即在 (k_x, k_y) 方向上的。这样在旋转后的吸收层中，σ_x 描述的是在 (k_x, k_y) 方向上的伸缩尺度，而不是在原来 x 方向上的。

或者考虑在圆的计算区域中，如图3所示

但在圆中给的吸收层参数需要沿着径向 r 方向上给出，而不是在 (k_x, k_y) 方向上的。可以将极坐标形式下的 Maxwell's Equations 转换到笛卡尔坐标系下，对应的 Jacobian 矩阵为

$$\mathcal{P} = \begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & 0 \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -r\sin\theta & 0 \\ \sin\theta & r\cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{5.47}$$

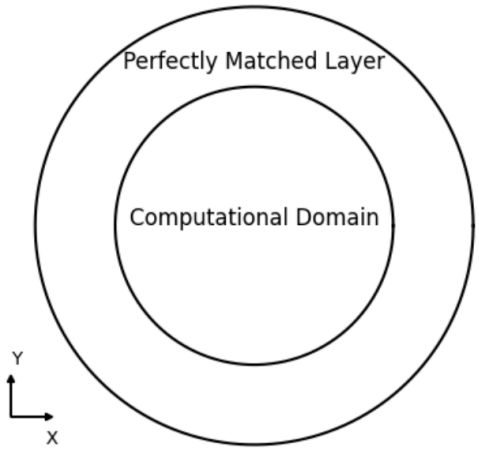


图 3: 在圆的情形下吸收边界层 PML_circle

在此基础上做一个伸缩的旋转变换

$$\mathcal{R} = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\frac{1}{r}\sin\theta & \frac{1}{r}\cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{5.48}$$

将这些变换拼凑起来

$$\mathcal{Q} = \mathcal{J}\mathcal{R}\mathcal{P} \tag{5.49}$$

这样便有

$$\frac{\mathcal{Q}\boldsymbol{\epsilon}\mathcal{Q}^T}{\det\mathcal{Q}} = \epsilon_a \frac{\mathcal{J}\mathcal{J}^T}{\det\mathcal{J}} \tag{5.50}$$

注意其中

$$\mathcal{J} = \begin{bmatrix} \frac{1}{s_r} & 0 & 0 \\ 0 & \frac{1}{s_\theta} & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{s_r} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

在坐标变换之后，还是使用 s_r 吗？对应于之前所说的问题。

6 Stability Analysis in 3D

在文章 [7] 中处理了 3-d second order wave equaiton，也同样发现了不稳定的因素，也是在处理 pml 的吸收参数中遇到了负值，可能可以模仿其方法进行处理。

由于散度算子在二维情形下可能有些关系不太明显，直接从三维的情形进行考虑，

$$\begin{aligned}\nabla\times\breve{\mathbf{H}}&=\left(j\omega\epsilon_0\epsilon_{r,\infty}\breve{\mathbf{E}}+j\omega\breve{\mathbf{P}}\right)\bar{\bar{s}},\\ \nabla\times\breve{\mathbf{E}}&=-j\omega\mu_0\mu_r\bar{\bar{s}}\breve{\mathbf{H}},\\ [1+(jw\tau)^{\alpha}]\,\breve{\mathbf{P}}&=\epsilon_0(\epsilon_{r,s}-\epsilon_{r,\infty})\breve{\mathbf{E}}.\end{aligned}\tag{6.1}$$

3D Maxwell in C

其中

$$\bar{\bar{s}}=\begin{bmatrix}\frac{s_ys_z}{s_x}&0&0\\0&\frac{s_xs_z}{s_y}&0\\0&0&\frac{s_xs_y}{s_z}\end{bmatrix}\tag{6.2}$$

且

$$s_i=1+\frac{\sigma_i}{j\omega\epsilon_0},\quad i=x,y,z\tag{6.3}$$

$$\begin{aligned}\frac{s_ys_z}{s_x}-1&=\frac{(1+\frac{\sigma_y}{j\omega\epsilon_0})(1+\frac{\sigma_z}{j\omega\epsilon_0})}{1+\frac{\sigma_x}{j\omega\epsilon_0}}-1\\&=\frac{\frac{\sigma_y\sigma_z}{(j\omega\epsilon_0)^2}+\frac{\sigma_y}{j\omega\epsilon_0}+\frac{\sigma_z}{j\omega\epsilon_0}-\frac{\sigma_x}{j\omega\epsilon_0}}{1+\frac{\sigma_x}{j\omega\epsilon_0}}\\&=\frac{\sigma_y\sigma_z+j\omega\epsilon_0(\sigma_y+\sigma_z-\sigma_x)}{(j\omega\epsilon_0)^2+j\omega\epsilon_0\sigma_x}\end{aligned}\tag{6.4}$$

对于方程组(6.1) 的第一个式子，写作分量的形式:

$$\frac{\partial H_z}{\partial y}-\frac{\partial H_y}{\partial z}=j\omega\epsilon_0\epsilon_{r,\infty}E_x+j\omega P_x+j\omega(\frac{s_ys_z}{s_x}-1)(\epsilon_0\epsilon_{r,\infty}E_x+P_x)\tag{6.5}$$

不妨记

$$\begin{aligned}(\frac{s_ys_z}{s_x}-1)E_x&=\frac{\sigma_y\sigma_z+j\omega\epsilon_0(\sigma_y+\sigma_z-\sigma_x)}{(j\omega\epsilon)^2+j\omega\epsilon\sigma_x}E_x=u_x,\\(\frac{s_ys_z}{s_x}-1)P_x&=\frac{\sigma_y\sigma_z+j\omega\epsilon_0(\sigma_y+\sigma_z-\sigma_x)}{(j\omega\epsilon)^2+j\omega\epsilon\sigma_x}P_x=v_x.\end{aligned}$$

将辅助变量对应的方程，转化到时域中，满足

$$\begin{aligned}\epsilon_0^2\frac{\partial^2u_x}{\partial t^2}+\epsilon_0\sigma_x\frac{\partial u_x}{\partial t}&=\sigma_y\sigma_zE_x+\epsilon_0(\sigma_y+\sigma_z-\sigma_x)\frac{\partial E_x}{\partial t},\\\epsilon_0^2\frac{\partial^2v_x}{\partial t^2}+\epsilon_0\sigma_x\frac{\partial v_x}{\partial t}&=\sigma_y\sigma_zP_x+\epsilon_0(\sigma_y+\sigma_z-\sigma_x)\frac{\partial P_x}{\partial t}\end{aligned}\tag{6.6}$$

再引入 u_x, v_x 对应的辅助变量 J_x, Q_x 将上述的二阶导数方程转化为一阶导数方程组。令

$$J_x = \frac{\partial u_x}{\partial t} - \frac{1}{\epsilon_0}(\sigma_y + \sigma_z - \sigma_x)E_x,$$
$$Q_x = \frac{\partial v_x}{\partial t} - \frac{1}{\epsilon_0}(\sigma_y + \sigma_z - \sigma_x)P_x.$$

相应的 J_x, Q_x 对应的方程为:

$$\begin{aligned}\frac{\partial J_x}{\partial t} &= -\frac{\sigma_x}{\epsilon_0}J_x + \frac{(\sigma_y - \sigma_x)(\sigma_z - \sigma_x)}{\epsilon_0^2}E_x,\\ \frac{\partial Q_x}{\partial t} &= -\frac{\sigma_x}{\epsilon_0}Q_x + \frac{(\sigma_y - \sigma_x)(\sigma_z - \sigma_x)}{\epsilon_0^2}P_x.\end{aligned} \tag{6.7}$$

同理，对于方程组(6.1) 的第二个式子，写做分量的形式，

$$\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} = -j\omega\mu_0\mu_r(\frac{s_x s_y}{s_z} - 1)H_x - j\omega\mu_0\mu_r H_x, \tag{6.8}$$

对于每一个分量，我们仍采用辅助变量的方式进行描述，设

$$(\frac{s_x s_y}{s_z} - 1)H_x = w_x, \tag{6.9}$$

w_x 满足的方程在时域下为

$$\epsilon_0^2\frac{\partial^2 w_x}{\partial t^2} + \epsilon_0\sigma_x\frac{\partial w_x}{\partial t} = \sigma_y\sigma_z H_x + \epsilon_0(\sigma_y + \sigma_z - \sigma_x)\frac{\partial H_x}{\partial t}. \tag{6.10}$$

引入辅助变量 $R_x = \dfrac{\partial w_x}{\partial t} - \frac{1}{\epsilon_0}(\sigma_y + \sigma_z - \sigma_x)w_x$，其对应的状态方程为

$$\frac{\partial R_x}{\partial t} = -\frac{\sigma_x}{\epsilon_0}R_x + \frac{(\sigma_y - \sigma_x)(\sigma_z - \sigma_x)}{\epsilon_0^2}H_x. \tag{6.11}$$

总结下来，3 维 Maxwell 方程组在 Cole-Cole 介质下表达式为

$$\begin{aligned}
\frac{\partial\left(\epsilon_0\epsilon_{r,\infty}\vec{E}+\vec{P}\right)}{\partial t} &= \nabla\times\vec{H}-\frac{1}{\epsilon_0}\cdot D_1\left(\epsilon_0\epsilon_{r,\infty}\vec{E}+\vec{P}\right)-\epsilon_0\epsilon_{r,\infty}\vec{J}-\vec{Q},\\
\frac{\partial\vec{J}}{\partial t} &= -\frac{1}{\epsilon_0}D_2\vec{J}+\frac{1}{\epsilon_0^2}D_3\vec{E},\\
\frac{\partial\vec{Q}}{\partial t} &= -\frac{1}{\epsilon_0}D_2\vec{Q}+\frac{1}{\epsilon_0^2}D_3\vec{P},\\
\mu_0\mu_r\frac{\partial\vec{H}}{\partial t} &= -\nabla\times\vec{E}-\frac{1}{\epsilon_0}D_1\vec{H}-\mu_0\mu_r\vec{R},\\
\frac{\partial\vec{R}}{\partial t} &= -\frac{1}{\epsilon_0}D_2\vec{R}+\frac{1}{\epsilon_0^2}D_3\vec{H},\\
\vec{P}+\tau^\alpha\frac{\partial^\alpha\vec{P}}{\partial t^\alpha} &= \epsilon_0\left(\epsilon_{r,s}-\epsilon_{r,\infty}\right)\vec{E},
\end{aligned} \tag{6.12}$$

其中

$$\begin{aligned}
D_1 &= \begin{pmatrix} \sigma_y+\sigma_z-\sigma_x & & \\ & \sigma_x+\sigma_z-\sigma_y & \\ & & \sigma_x+\sigma_y-\sigma_z \end{pmatrix}, D_2 = \begin{pmatrix} \sigma_x & & \\ & \sigma_y & \\ & & \sigma_z \end{pmatrix}\\
D_3 &= \begin{pmatrix} (\sigma_y-\sigma_x)(\sigma_z-\sigma_x) & & \\ & (\sigma_x-\sigma_y)(\sigma_z-\sigma_y) & \\ & & (\sigma_x-\sigma_z)(\sigma_y-\sigma_z) \end{pmatrix}.
\end{aligned} \tag{6.13}$$

按照此中系数矩阵的取法，仍然会存在非正的情况，按照 wave equation 中的处理方式还是存在着问题。

在三维情形下，可以用 coordinate 变换的方式来写 PML 系统的表示方式，参考 [7] 的形式，在 PML 层中的表示为

$$\frac{\partial}{\partial\tilde{x}}=\frac{1}{s_x}\frac{\partial}{\partial x},\quad \frac{\partial}{\partial\tilde{y}}=\frac{1}{s_y}\frac{\partial}{\partial y},\quad \frac{\partial}{\partial\tilde{z}}=\frac{1}{s_z}\frac{\partial}{\partial z}.$$

按照这样的方式推导出来的系统为

$$\frac{\partial\left(\epsilon_0\epsilon_{r,\infty}\vec{E}+\vec{P}\right)}{\partial t}=\nabla\times\vec{H}+\frac{1}{\epsilon_0}\nabla\times(D_1\vec{v})-\frac{1}{\epsilon_0}D_2\left(\epsilon_0\epsilon_{r,\infty}\vec{E}+\vec{P}\right)-\frac{1}{\epsilon_0^2}D_3\left(\epsilon_0\epsilon_{r,\infty}\vec{u}+\vec{w}\right), \tag{6.14}$$

$$\mu_0\mu_r\frac{\partial\vec{H}}{\partial t}=-\nabla\times\vec{E}-\frac{1}{\epsilon_0}\nabla\times(D_1\vec{u})-\frac{\mu_0\mu_r}{\epsilon_0}D_2\vec{H}-\frac{\mu_0\mu_r}{\epsilon_0^2}D_3\vec{v}, \tag{6.15}$$

$$\frac{\partial\vec{u}}{\partial t}=\vec{E}, \tag{6.16}$$

$$\frac{\partial\vec{v}}{\partial t}=\vec{H}, \tag{6.17}$$

$$\frac{\partial\vec{w}}{\partial t}=\vec{P}, \tag{6.18}$$

$$\vec{P}+\tau_0^\alpha\int_0^\infty\vec{\varphi}d\lambda=\epsilon_0(\epsilon_{r,s}-\epsilon_{r,\infty})\vec{E}, \tag{6.19}$$

$$\frac{\partial\vec{\varphi}}{\partial t}=-\lambda\vec{\varphi}+\frac{\sin\pi\alpha}{\pi}\lambda^{\alpha-1}\frac{\partial\vec{P}}{\partial t}, \tag{6.20}$$

其中，

$$D_1=\begin{pmatrix}\sigma_x & & \\ & \sigma_y & \\ & & \sigma_z\end{pmatrix},D_2=\begin{pmatrix}\sigma_y+\sigma_z & & \\ & \sigma_z+\sigma_x & \\ & & \sigma_x+\sigma_y\end{pmatrix},D_3=\begin{pmatrix}\sigma_y\sigma_z & & \\ & \sigma_z\sigma_x & \\ & & \sigma_x\sigma_y\end{pmatrix}. \tag{6.21}$$

首先看空间导数项：记

$$\vec{E}^*=\vec{E}+\frac{1}{\epsilon_0}D_1\vec{u},\quad \vec{H}^*=\vec{H}+\frac{1}{\epsilon_0}D_1\vec{v},$$

第一个方程，乘上 $\vec{E}^*$，第二个方程组，乘上 $\vec{H}^*$，两式相加，则关于空间导数的部分：

$$\int_\Omega\vec{E}^*\cdot\left(\nabla\times\vec{H}^*\right)-\vec{H}^*\cdot\left(\nabla\times\vec{E}^*\right)dV=-\int_\Omega\nabla\cdot\left(\vec{E}^*\times\vec{H}^*\right)dV \tag{6.22}$$

$$=-\int_{\partial\Omega}\left(\vec{E}^*\times\vec{H}^*\right)\cdot\vec{n}dS=-\int_{\partial\Omega}\left(\vec{n}\times\vec{E}^*\right)\cdot\vec{H}^*dS. \tag{6.23}$$

在完美导电边界条件下 $\vec{n}\times\vec{E}=0$，初值条件 $\vec{u}(\boldsymbol{x},0)=0$，则 $\vec{n}\times\vec{E}=\vec{n}\times\partial_t\vec{u}=\partial_t\left(\vec{n}\times\vec{u}\right)=0$，则 $\vec{n}\times\vec{u}=0$。

当计算区域的单位外法向量 $\vec{n}$ 平行与坐标轴时，即只有一个分量，如 $\vec{n}=(1,0,0)^T$，则

$$\vec{n}\times\vec{u}=\begin{pmatrix}0\\u_z\\-u_y\end{pmatrix}=0$$

则 $D_1\vec{u}=(\sigma_x u_x,0,0)^T$，则 $\vec{n}\times(D_1\vec{u})=0$。那么

$$\vec{n}\times\vec{E}^*=\vec{n}\times\left(\vec{E}+\frac{1}{\epsilon_0}D_1\vec{u}\right)=\vec{n}\times\vec{E}+\frac{1}{\epsilon_0}\vec{n}\times(D_1\vec{u})=0,$$

所以在此种测试函数和初值条件的选取方法下，完美导电边界条件使得空间导数项的贡献为零。

6.1 CFS-PML 尝试

在三维情形下，如果采用 CFS-PML 进行建模的话会是怎么样的结果。CFS 表示复频移的坐标伸缩，coordinate 的变换形式仍然写做

$$\frac{\partial}{\partial\tilde{x}}=\frac{1}{s_x}\frac{\partial}{\partial x},\quad \frac{\partial}{\partial\tilde{y}}=\frac{1}{s_y}\frac{\partial}{\partial y},\quad \frac{\partial}{\partial\tilde{z}}=\frac{1}{s_z}\frac{\partial}{\partial z},$$

其中

$$s_i=\kappa_i+\frac{\sigma_i}{\alpha_i+jw\epsilon_0},\quad i=x,y,z. \tag{6.24}$$

一般的 $\kappa_i=1$, $\alpha_i\geq 0$.

考虑二维情形，频域下的 Maxwell 方程组为：

$$\frac{\partial\breve{H}_y}{\partial x}-\frac{\partial\breve{H}_x}{\partial y}=j\omega\epsilon_0\epsilon_{r,\infty}\breve{E}_z+j\omega\breve{P}, \tag{6.25a}$$

$$\frac{\partial\breve{E}_z}{\partial y}=-j\omega\mu_0\mu_r\breve{H}_x, \tag{6.25b}$$

$$-\frac{\partial\breve{E}_z}{\partial x}=-j\omega\mu_0\mu_r\breve{H}_y, \tag{6.25c}$$

$$\breve{P}+(j\omega\tau_0)^\alpha\breve{P}=\epsilon_0(\epsilon_{r,s}-\epsilon_{r,\infty})\breve{E}_z. \tag{6.25d}$$

那么在这样的坐标伸缩变换下，相应的含有 PML 的方程组写做：

$$\frac{1}{s_x}\frac{\partial \breve{H}_y}{\partial x} - \frac{1}{s_y}\frac{\partial \breve{H}_x}{\partial y} = j\omega\epsilon_0\epsilon_{r,\infty}\breve{E}_z + j\omega\breve{P}_z, \tag{6.26a}$$

2D Maxwell in C

$$\frac{1}{s_y}\frac{\partial \breve{E}_z}{\partial y} = -j\omega\mu_0\mu_r\breve{H}_x, \tag{6.26b}$$

2D Maxwell in C

$$-\frac{1}{s_x}\frac{\partial \breve{E}_z}{\partial x} = -j\omega\mu_0\mu_r\breve{H}_y, \tag{6.26c}$$

2D Maxwell in C

$$\breve{P}_z + (j\omega\tau_0)^\alpha\breve{P}_z = \epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty})\breve{E}_z. \tag{6.26d}$$

2D Maxwell in C

equency CFS-PML

在

6.2 长时间稳定性

考虑在静默状态下的系统稳定性。即当空间导数项均为 0 ，参见 [1]，同时也考虑最简单的 Debye 色散介质，系统变为

$$\frac{\partial\left(\epsilon_0\epsilon_{r,\infty}\vec{E} + \vec{P}\right)}{\partial t} = -\frac{1}{\epsilon_0}D_2\left(\epsilon_0\epsilon_{r,\infty}\vec{E} + \vec{P}\right) - \frac{1}{\epsilon_0^2}D_3\left(\epsilon_0\epsilon_{r,\infty}\vec{u} + \vec{w}\right), \tag{6.27}$$

$$\mu_0\mu_r\frac{\partial\vec{H}}{\partial t} = -\frac{\mu_0\mu_r}{\epsilon_0}D_2\vec{H} - \frac{\mu_0\mu_r}{\epsilon_0^2}D_3\vec{v}, \tag{6.28}$$

$$\frac{\partial\vec{u}}{\partial t} = \vec{E}, \tag{6.29}$$

$$\frac{\partial\vec{v}}{\partial t} = \vec{H}, \tag{6.30}$$

$$\frac{\partial\vec{w}}{\partial t} = \vec{P}, \tag{6.31}$$

$$\vec{P} + \frac{\partial\vec{P}}{\partial t} = \epsilon_0(\epsilon_{r,s} - \epsilon_{r,\infty})\vec{E}, \tag{6.32}$$

首先对于 $\left(\vec{H}, \vec{v}\right)$ 构成的系统，归一化后，写做

$$\frac{\partial}{\partial t}\begin{pmatrix}\vec{H}\\ \vec{v}\end{pmatrix} = \begin{pmatrix}-D_2 & -D_3\\ I & 0\end{pmatrix}\begin{pmatrix}\vec{H}\\ \vec{v}\end{pmatrix}, \tag{6.33}$$

eq: eigen H v

如果假设 C 和 D 可交换（$CD = DC$）：

$$\begin{pmatrix}A & B\\ C & D\end{pmatrix}\begin{pmatrix}D & 0\\ -C & I\end{pmatrix} = \begin{pmatrix}AD - BC & B\\ CD - DC & D\end{pmatrix} = \begin{pmatrix}AD - BC & B\\ 0 & D\end{pmatrix}$$

现在两边取行列式：

$$\det(M)\cdot\det(D)=\det(AD-BC)\cdot\det(D)$$

如果 $\det(D)\neq 0$，消去 $\det(D)$ 即得证。如果 $\det(D)=0$，可以通过连续性参数微扰证明。当 C,D 可交换时，则有

$$\det\begin{pmatrix}A & B\\ C & D\end{pmatrix}=\det(AD-CB).$$

对于方程组 (6.33) 的系数矩阵，对应的特征值方程为

$$\det\begin{pmatrix}-D_2-\Lambda I & -D_3\\ I & -\Lambda I\end{pmatrix}=0,$$

因为 D_2 和 D_3 均为对角矩阵，所以有

$$\det((-\Lambda I)(-D_2-\Lambda I)-(I)(-D_3))=\det(\Lambda D_2+\Lambda^2 I+D_3)=0,$$

带入 D_2,D_3 的表达式，相应的特征根为

$$\{-\sigma_x,-\sigma_x,-\sigma_y,-\sigma_y,-\sigma_z,-\sigma_z\}.$$

当 $\sigma_x=\sigma_z=0$ 时，系统对应的系数矩阵为

$$\frac{\partial}{\partial t}\begin{pmatrix}H_x\\ H_y\\ H_z\\ v_x\\ v_y\\ v_z\end{pmatrix}=\begin{pmatrix}-\sigma_y & & & 0 & & \\ & 0 & & & 0 & \\ & & -\sigma_y & & & 0\\ 1 & & & & & \\ & 1 & & & & \\ & & 1 & & & \end{pmatrix}\begin{pmatrix}H_x\\ H_y\\ H_z\\ v_x\\ v_y\\ v_z\end{pmatrix},$$

由于矩阵是稀疏且分块解耦的，我们可以针对 x,y,z 三对分量 (H_i,v_i) 分别看：

对于 x 分量 (H_x,v_x) 和 z 分量 (H_z,v_z)：子矩阵为：

$$\begin{pmatrix}-\sigma_y & 0\\ 1 & 0\end{pmatrix}$$

特征方程：$\lambda(-\sigma_y-\lambda)=0\implies\lambda_1=0,\lambda_2=-\sigma_y$。那么：这是两个互异的特征值（假设 $\sigma_y\neq 0$），矩阵可对角化，解是有界的。

对于 y 分量 (H_y, v_y)：子矩阵为：

$$\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

特征方程：$\lambda^2 = 0 \implies \lambda_1 = \lambda_2 = 0$。此为二重 0 特征根，且对应一个非零的 Jordan 块（右上角为0，左下角为1，具有幂零性质）。

由于 y 分量对应的子矩阵是 Jordan 块 $\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$，其动力学方程为：

$$\frac{\partial H_y}{\partial t} = 0$$

$$\frac{\partial v_y}{\partial t} = H_y$$

解的表达式：

$$H_y(t) = H_y(0)$$

$$v_y(t) = H_y(0) \cdot t + v_y(0)$$

注：这是该子系统的不稳定（线性增长）来源。

综上所述，在 $\sigma_x = \sigma_z = 0$ 的条件下，系统解的显式表达为：

$$\vec{H}(t) = \begin{pmatrix} H_x(0)e^{-\sigma_y t} \\ H_y(0) \\ H_z(0)e^{-\sigma_y t} \end{pmatrix},$$

$$\vec{v}(t) = \begin{pmatrix} v_x(0) + \frac{H_x(0)}{\sigma_y}(1 - e^{-\sigma_y t}) \\ \mathbf{H_y(0)} \cdot \mathbf{t} + v_y(0) \\ v_z(0) + \frac{H_z(0)}{\sigma_z}(1 - e^{-\sigma_y t}) \end{pmatrix},$$

这导致了不稳定现象。而且 $\vec{v}(t)$ 的第二项随着时间的发展，增长到正无穷，这导致无法取到相应的能量泛函

类似的，对于

$$\left(\vec{E}, \vec{P}, \vec{u}, \vec{w}\right)$$

构成的系统，归一化后，写做

$$\frac{\partial}{\partial t}\begin{pmatrix}\vec{E}\\ \vec{P}\\ \vec{u}\\ \vec{w}\end{pmatrix}=\begin{pmatrix}-D_2 & -D_2 & -D_3 & -D_3\\ I & -I & 0 & 0\\ I & 0 & 0 & 0\\ 0 & I & 0 & 0\end{pmatrix}\begin{pmatrix}\vec{E}\\ \vec{P}\\ \vec{u}\\ \vec{w}\end{pmatrix} \tag{6.34}$$

eq: eigen E u P

对应的特征矩阵为

$$M-\Lambda I=\left(\begin{array}{cc|cc}-D_2-\Lambda & -D_2 & -D_3 & -D_3\\ I & -I-\Lambda & 0 & 0\\ \hline I & 0 & -\Lambda & 0\\ 0 & I & 0 & -\Lambda\end{array}\right)=\begin{pmatrix}\mathbf{A} & \mathbf{B}\\ \mathbf{C} & \mathbf{D}\end{pmatrix},$$

由上面证明到的性质，$\mathbf{C}$ 是单位矩阵，和任意的矩阵 $\mathbf{D}$ 都是可交换的。所以有

$$\det(M-\Lambda I)=\det(\mathbf{AD}-\mathbf{BC}).$$

其中

$$\begin{aligned}\mathbf{AD}-\mathbf{BC}&=\begin{pmatrix}(-D_2-\Lambda)(-\Lambda)-(-D_3) & (-D_2)(-\Lambda)-(-D_3)\\ I(-\Lambda)-0 & (-I-\Lambda)(-\Lambda)-0\end{pmatrix}\\&=\begin{pmatrix}\Lambda D_2+\Lambda^2 I+D_3 & \Lambda D_2+D_3\\ -\Lambda I & \Lambda I+\Lambda^2 I\end{pmatrix},\end{aligned}$$

则由于 ΛI 是对角矩阵，那么

$$\begin{aligned}\det(M-\Lambda I)&=\det((\Lambda I+\Lambda^2 I)(\Lambda D_2+\Lambda^2 I+D_3)-(-\Lambda I)(\Lambda D_2+D_3))\\&=\det(\Lambda\left(\Lambda^3 I+\Lambda^2\left(I+D_2\right)+\Lambda\left(2D_2+D_3\right)+2D_3\right)).\end{aligned}$$

则对应的一个特征根为 $\Lambda=0$，其他的特征根满足

$$\det\left(\Lambda^3 I+\Lambda^2\left(I+D_2\right)+\Lambda\left(2D_2+D_3\right)+2D_3\right)=0.$$

由于 D_2, D_3 均为对角矩阵，所以有

$$\begin{aligned}\lambda^3+(1+\sigma_y+\sigma_z)\lambda^2+(2\sigma_y+2\sigma_z+\sigma_y\sigma_z)\lambda+2\sigma_y\sigma_z=0,\\ \lambda^3+(1+\sigma_z+\sigma_x)\lambda^2+(2\sigma_z+2\sigma_x+\sigma_z\sigma_x)\lambda+2\sigma_z\sigma_x=0,\\ \lambda^3+(1+\sigma_x+\sigma_y)\lambda^2+(2\sigma_x+2\sigma_y+\sigma_x\sigma_y)\lambda+2\sigma_x\sigma_y=0.\end{aligned}$$

判断这样的特征方程是否有负的实部。利用 Routh-Hurwitz 判据：

- 每一项的系数 $a_n > 0$ 均为正。若有 $\sigma_y = 0$，则第一个式子退化为

$$\lambda^3 + (1+\sigma_z)\lambda^2 + 2\sigma_z\lambda = 0,$$

对应的根为

$$\lambda = 0, \quad \lambda = \frac{-(1+\sigma_z) \pm \sqrt{(1+\sigma_z)^2 - 8\sigma_z}}{2},$$

根的实部必然是小于0 的。若 $\sigma_z = 0$，则

$$\lambda_{1,2} = 0, \quad \lambda_3 = -1,$$

- 每一项的系数 $a_3 = 1$, $a_2 = 1 + \sigma_y + \sigma_z$, $a_1 = 2\sigma_y + 2\sigma_z + \sigma_y\sigma_z$, $a_0 = 2\sigma_y\sigma_z$, 需要满足 $a_2a_1 > a_3a_0$ (根据上一点分析，就只需要考虑 $\sigma_{y,z} > 0$ 的情形)

$$\begin{aligned} a_2a_1 - a_3a_0 &= 2(\sigma_y + \sigma_z) + 2(\sigma_y + \sigma_z)^2 + \sigma_y\sigma_z(\sigma_y + \sigma_z) - \sigma_y\sigma_z \\ &> 2(\sigma_y + \sigma_z) + 2(\sigma_y - \sigma_z)^2 + \sigma_y\sigma_z(\sigma_y + \sigma_z) > 0, \end{aligned}$$

故根据判定准则，此三阶方程的根的实部均为非正的。综上所述，对于长时间的行为，有如下的结论：

当位于 $\sigma_y = \sigma_z = 0$ 的区域中，0 特征值为三重根，其他的特征值实部均为负值；对应的系统矩阵为：

$$M = \begin{pmatrix} 0 & 0 & 0 & 0 \\ I & -I & 0 & 0 \\ I & 0 & 0 & 0 \\ 0 & I & 0 & 0 \end{pmatrix}.$$

6.3 T 时间内单调递减，超过这个时间不稳定?

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